

《Automotive Configuration》

Chapter 10

Lecturer : Dr. HU Zongjie / Associate Professor

Contact: zongjie-hu@tongji.edu.cn

Chapter 10 Engine Ignition System

第十章 发动机点火系统

10.1 Overview

概述

10.2 Components and Fundamental of Conventional Ignition System

传统点火系统的组成与工作原理

10.3 Electronic Ignition System

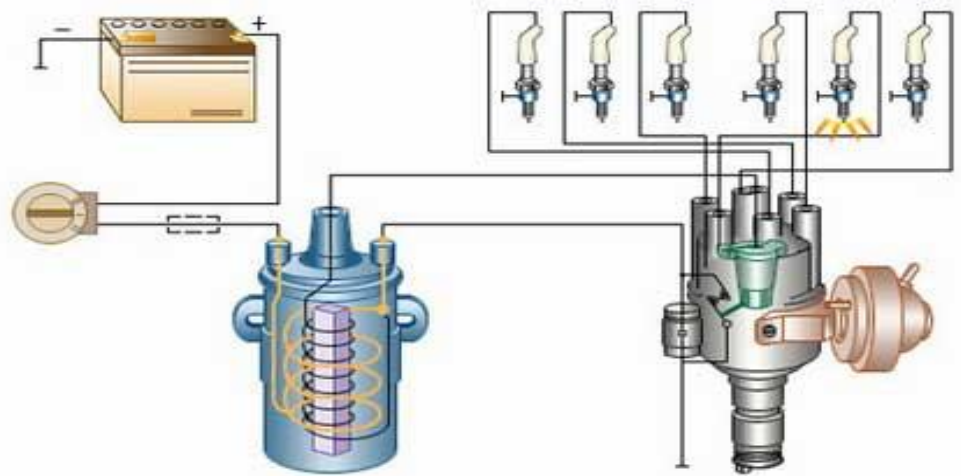
电子点火系统

10.4 Microcomputer Controlled Ignition System

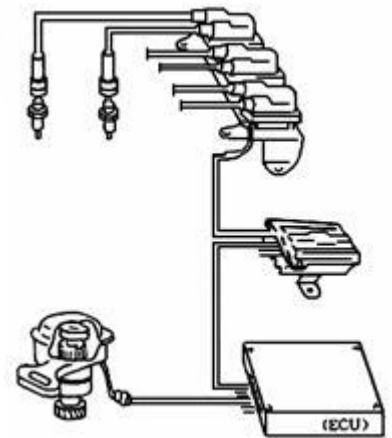
微机控制点火系统

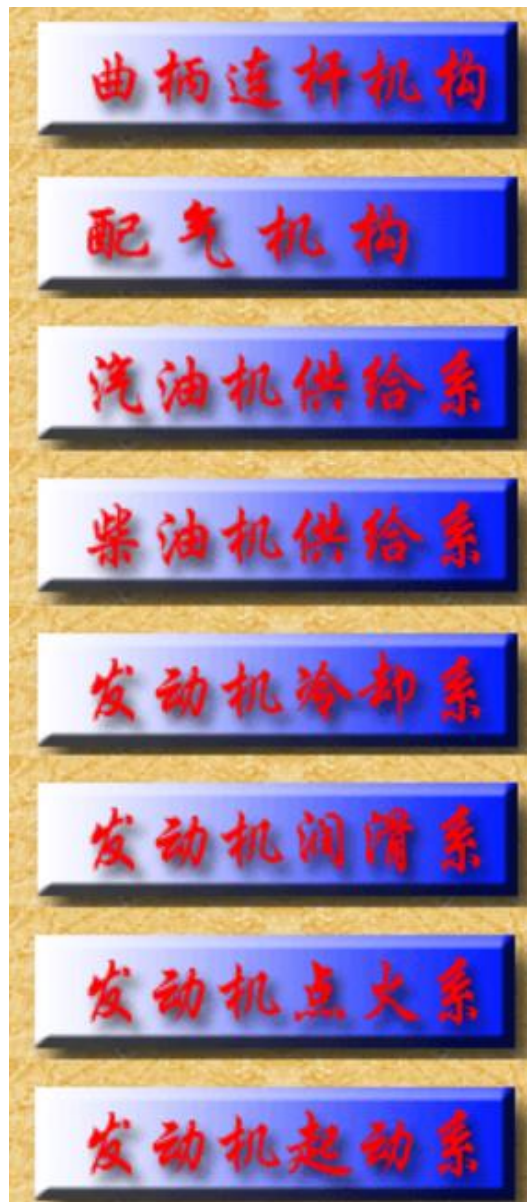
10.5 Automotive Power Supply

汽车电源



点火系统





Two mechanisms

Five systems

**Suppling system –
Fuel & Air**

**Dedicated for
Gasoline engines**

§ 10.1 Overview

1 Function(功用)

Make two electrodes (电极) of spark plug(火花塞) generate a strong enough electric spark at time, to ignite the combustible mixture according to the engine cylinder ignition order, so as to do work .

2 Classification(分类) and main components(主要部件)

① Battery-operated ignition system (蓄电池点火系统) :

[Battery(蓄电池)、 generator/alternator(发电机)、 ignition coil(点火线圈)]、 breaker (断电器)

② Electronic ignition system(电子点火系统):

[]、 triode(三极管)

③ Microcomputer control ignition system(微机控制点火系统) :

[]、 microcomputer(微机)

④ Single Ignition System(单体点火系统)

⑤ Magneto ignition system (磁电机点火系统) : produce high voltage by magneto systems directly.

3 Requirements on Ignition Sytem of Automotive Engine(汽车发动机对点火系的要求)

① High enough Breakdown voltage(击穿电压): 15~20kV

Factors : mixture, pressure, temperature and distance between two electrodes **lean / ultra-lean burn**

② High enough energy of spark

Normal operation(正常工作): 1~5mJ;

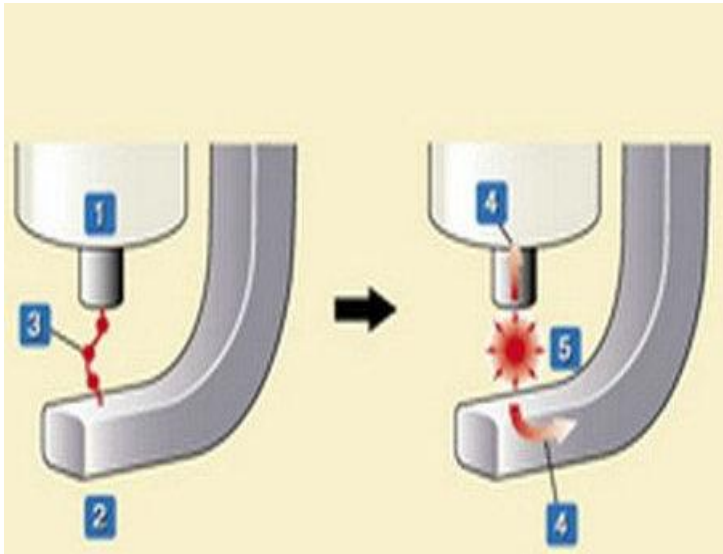
Idle speed (怠速): 50 ~ 80mJ;

Cold start: >100mJ.

≈1000mJ, torch ignition / jet ignition

③ Ignition timing(点火时刻) should be adjusted to the operating condition(工况).

CA-based, engine speed, knock control



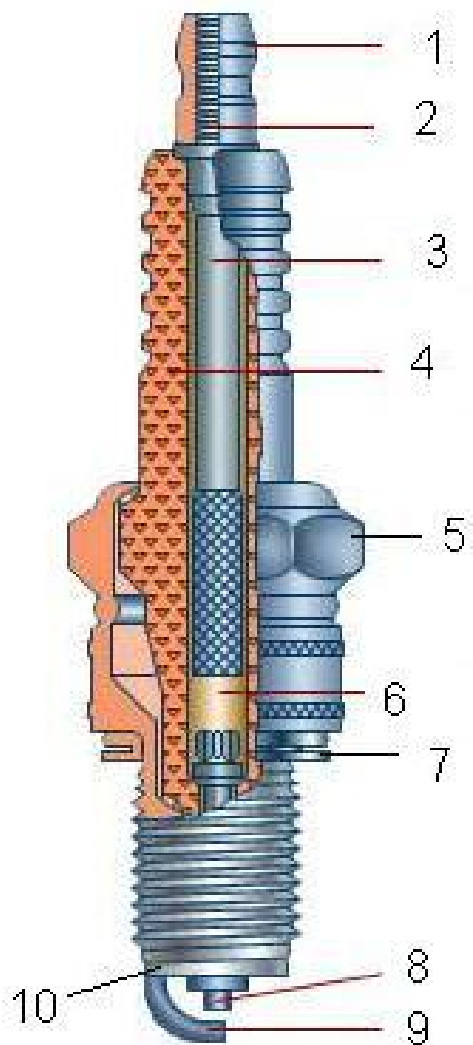
火花塞 (Spark Plug)

Function:

Import the pulse(脉冲) produced by ignition coil into the combustion chamber(燃烧室), generate electric spark between two electrodes, so as to ignite combustible mixture.



Structure of spark plug(火花塞构造)



1.terminal stud (接线螺柱)

2.connection thread (连接螺纹)

3.terminal stud(螺柱)

4.insulator (绝缘体)

5.shell (外壳)

6.special conductive seal (密封剂)

7.seal washer (密封垫圈)

8.center electrode (中心电极)

9.ground electrode (侧电极)

10. insulator skirt section (绝缘体裙部)

2) Ignition coil (点火线圈)

Function: transform low voltage into high voltage——step-up transformer (升压变压器) by electromagnetic induction principle

Types(分类):

◆ Open magnetic circuit
ignition coil

(开磁路点火线圈)

◆ Closed magnetic circuit
ignition coil

(闭磁路点火线圈)



The structure of ignition coil

1. high voltage coil joint (高压线接头)
2. added resistance joint (附加电阻接头)
3. iron core (铁心)
4. primary winding (初级绕组)
5. secondary winding (次级绕组)
6. shell (外壳)
7. top cover (上盖)
8. primary binding post (初级接线柱)

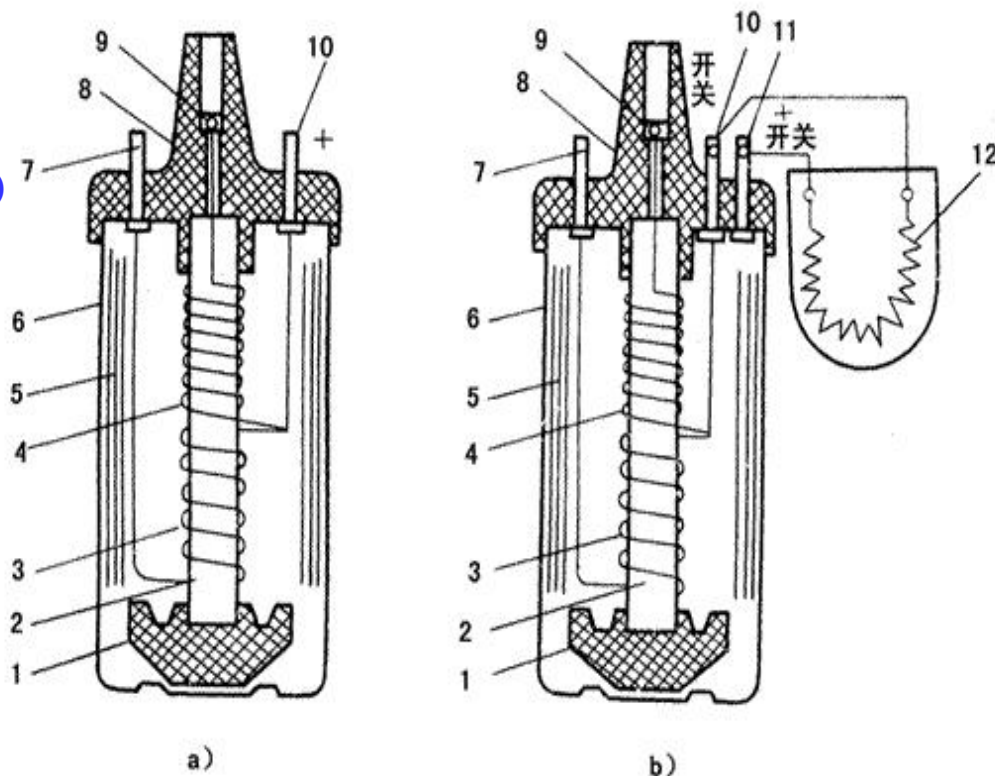
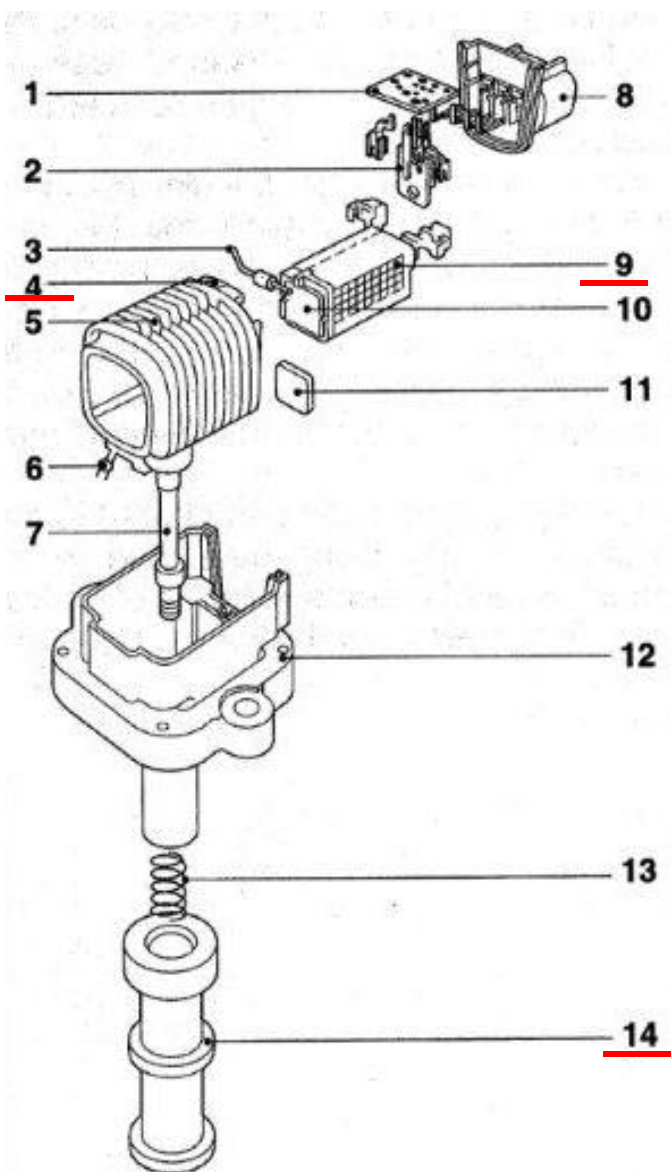


图 4-15 点火线圈结构示意图

1-瓷杯 2-铁心 3-初级绕组 4-次级绕组 5-钢片 6-外壳 7、“-”接柱
8-胶木盖 9-高压线插座 10-“+”或“开关”接柱 11-“开关”接柱 12-附加电阻





Design of the compact ignition coil
 1 Printed-circuit board, 2 Driver stage,
 3 Activation arc diode, 4 Secondary winding housing, 5 Secondary winding, 6 Contact plate, 7 High-voltage pin (connection to the contact spring), 8 Primary connector, 9 Primary winding, 10 I-core, 11 Permanent magnet, 12 O-core, 13 Spring (spark-plug contact), 14 Silicone sleeve (high-voltage insulation).

Microcomputer Controlled Distributorless Ignition system

无分电器的微机控制点火系统

Composition:

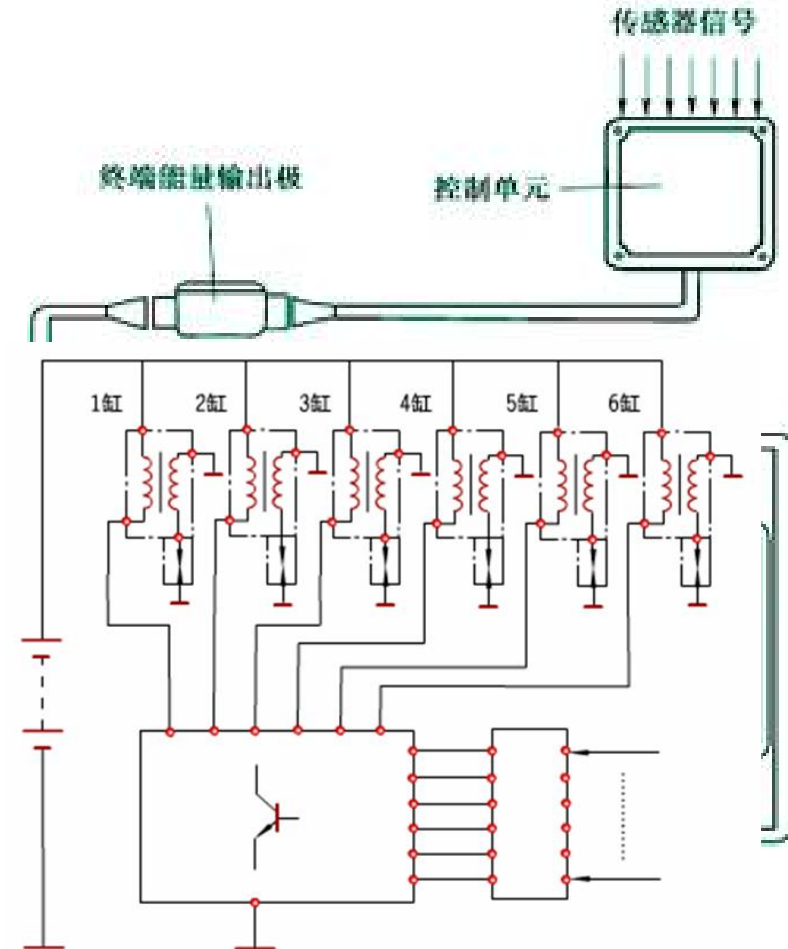
Low voltage power supply (低压电源), ignition switches, micro control unit (ECU), ignition coil, ignition controller, spark plugs, wire and sensors.

Types:

Independent ignition(独立点火) and simultaneous ignition(同时点火)

Independent ignition(独立点火方式):

Each cylinder is assigned an ignition coil which is installed on the top of the spark plug directly .



点火线圈独立、共用一个点火控制器的点火系统工作原理图

Automotive Power Supply (汽车电源)

Composition:

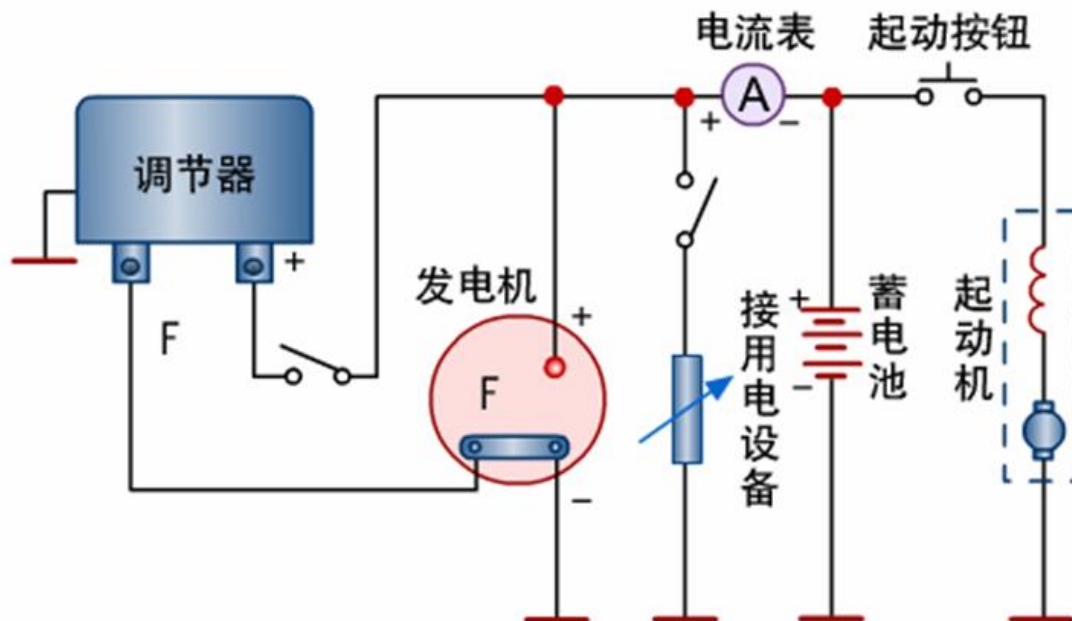
Battery(蓄电池)、Electric Generator(发电机)、voltage regulator(调节器)。

3 states:

Start and low speed;

Normal operation;

Big power consumption.

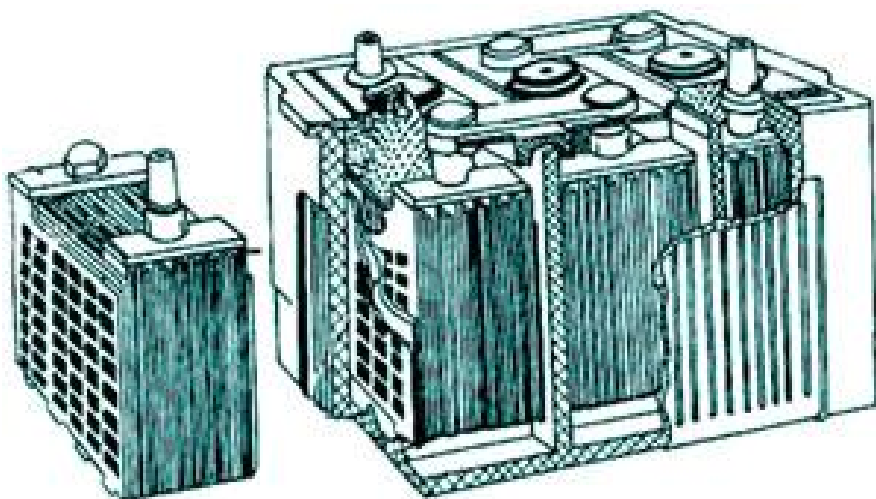


汽车电源电路

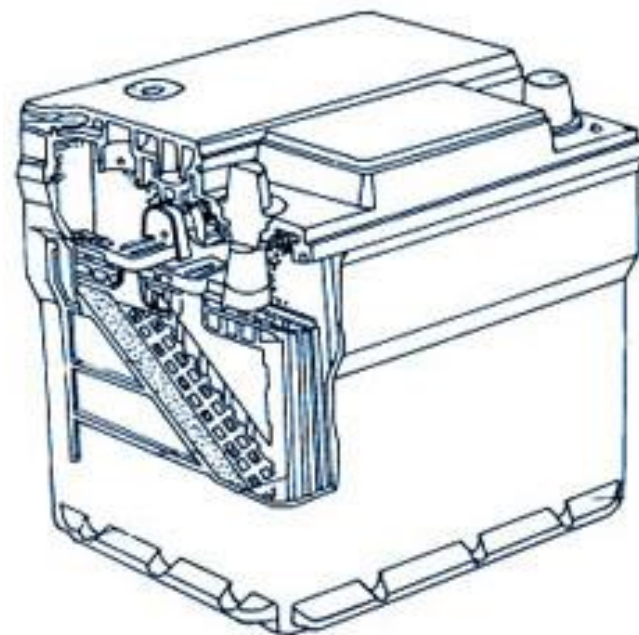
1 Battery(蓄电池)

◆Types: Alkaline and acidic Battery(碱性和酸性蓄电池).

◆Function: Generate power; Store electricity(储存电能);Stabilize the grid voltage(稳定电网电压).



蓄电池



免维护蓄电池的结构图

① Structure of Lead battery (铅蓄电池结构)

3 or 6 single cells , the single voltage is about 2 v.

3或6只单格电池串联组成，单格电压2V左右

Auto power : 12 v, 24 v or 6 v;

Electrolyte(电解液);

Electrolyte-adding hole(加液孔), air vent(通气孔)

② Capacity of the battery (蓄电池的容量)

20 hours rated capacity(20小时额定容量) (C20), unit : Ah;

rated Reserve capacity (Cm)(额定储备容量),unit: min;

cold start capability, Is(冷起动容量) unit:A

2 Electric Generator(发电机)

Requirement:

- ◆ Output current is DC(直流);
- ◆ Constant terminal voltage(端电压恒定), battery voltage $\leq$ output voltage $\leq$ electrical equipment permissible voltage;
- ◆ In high speed, the quantity of electricity can meet most of the electricity equipment needs.

Types of the silicon rectifying generator based on overall structure

- ◆ Conventional AC generator (普通交流发电机);
- ◆ Integrate alternator (整体式交流发电机);
- ◆ Brushless alternator (无刷交流发电机)
- ◆ Permanent magnet alternator (永磁交流发电机).

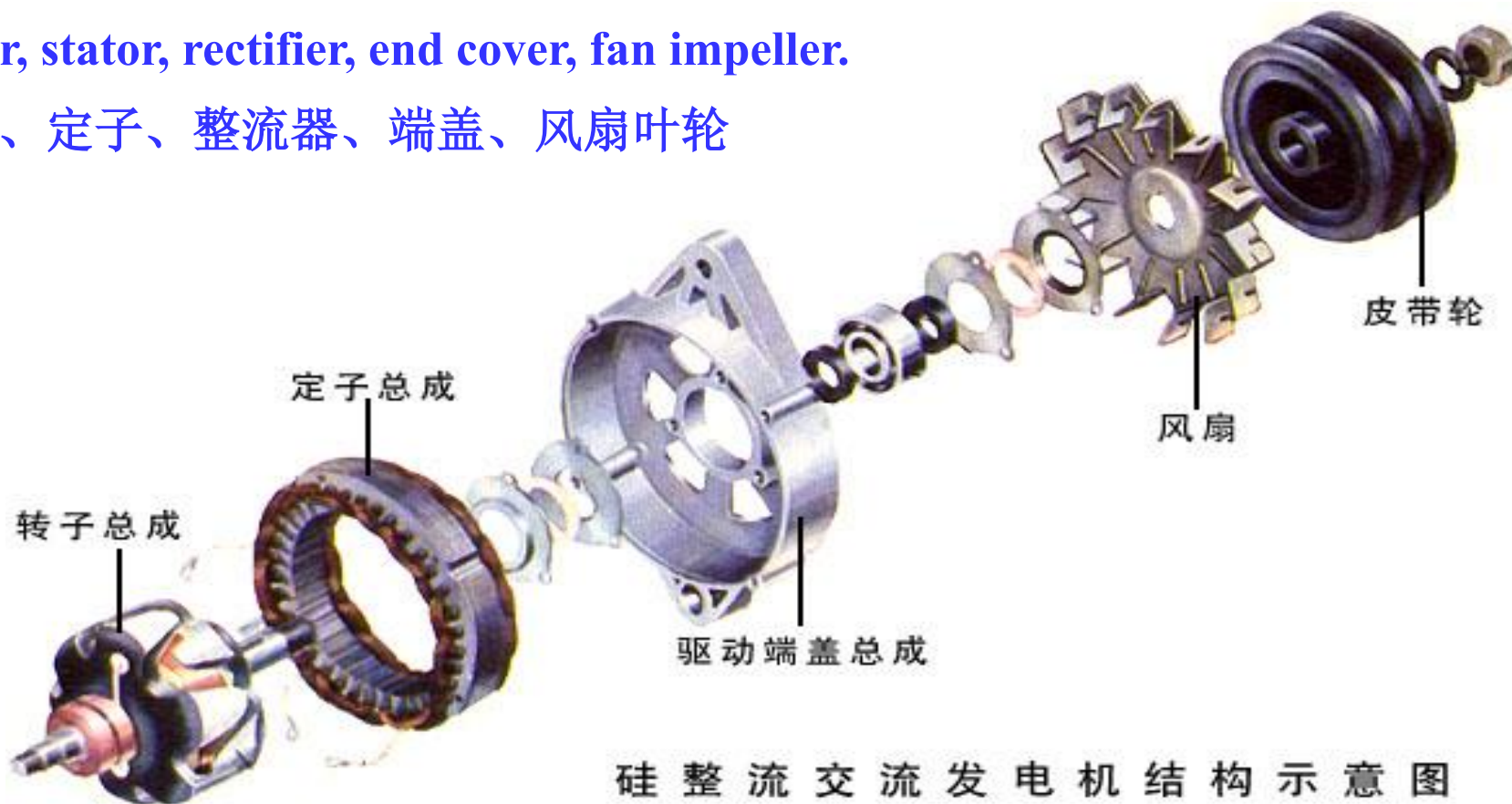
Structure of the silicon rectification alternator

(硅整流交流发电机的结构)

◆composition

Rotor, stator, rectifier, end cover, fan impeller.

转子、定子、整流器、端盖、风扇叶轮



硅整流交流发电机结构示意图

4 The Features of Ignition Sytem(点火系的特点)

- ① Single wire connectting to different electrical equipments(单线制联结)
- ② Negative Electrode connected to GND(负极搭铁)
- ③ Central electrode is set to be the negative electrode



5 Types of Engine Ignition System 发动机点火系的发展

◆ The simplest one —— magneto system

(磁电式点火系统)

◆ The first one using coil ——conventional coil system

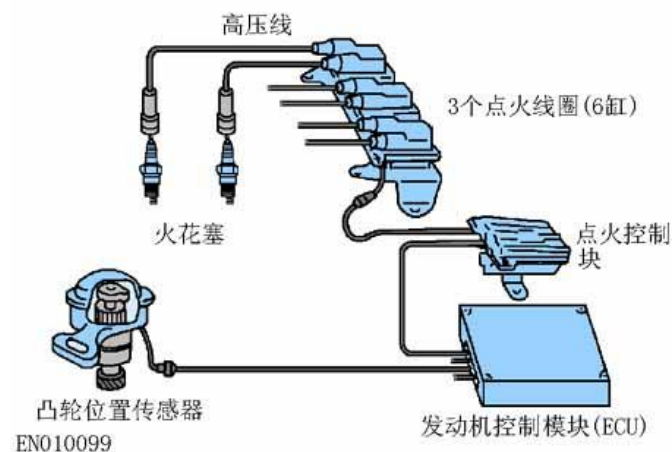
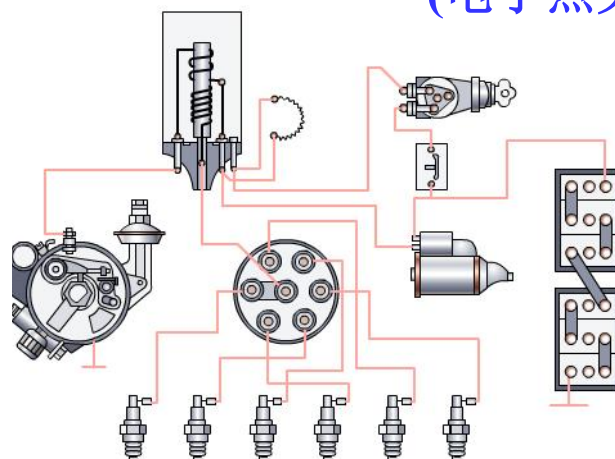
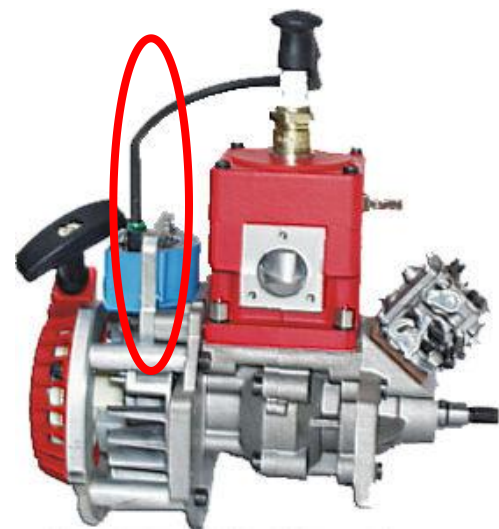
(传统点火系统)

◆ The most common used one—— semi-conductor ingnition sytem

(半导体点火系统)

◆ The latest one —— electronic ignition system

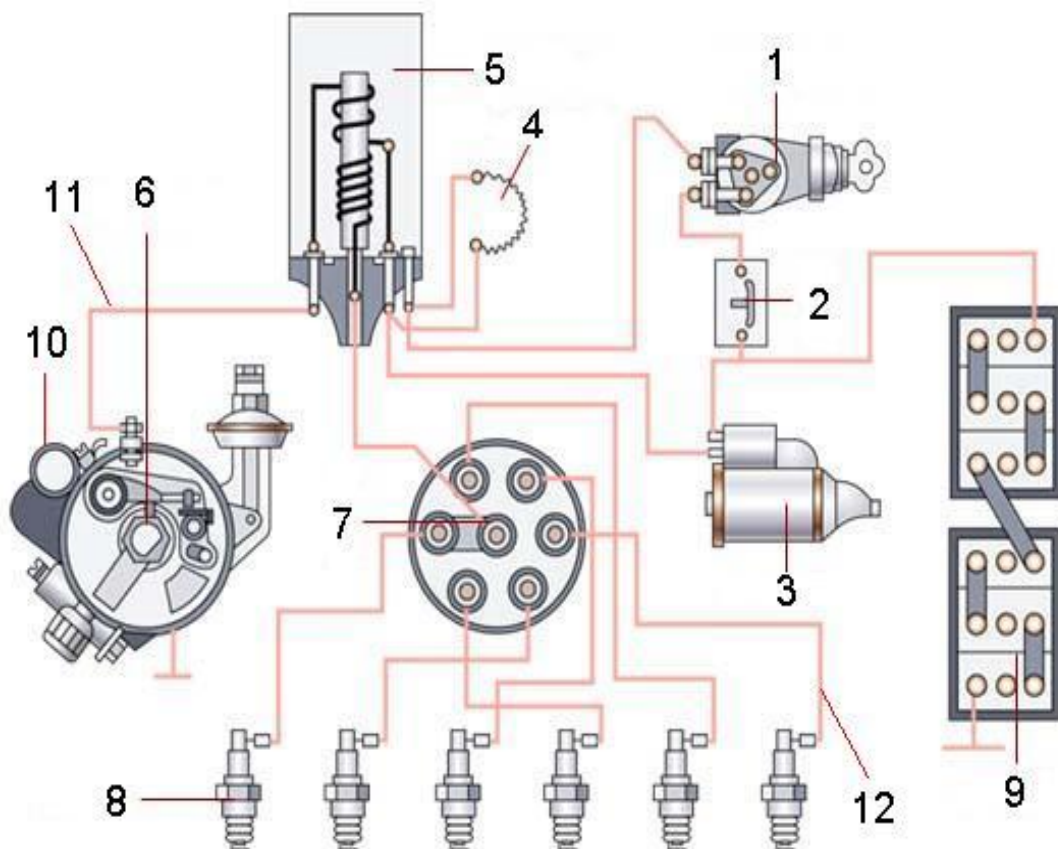
(电子点火系统)



§ 10.2 Components and Fundamental of Conventional Ignition System

传统点火系统组成与工作原理

Definition: use the battery and generator as power supply, rely on the ignition coil and distributor block(分电器) to ignite mixture successfully.



1. Ignition Switch (点火开关)

2. Ammeter (电流表)

3. Selfstarting Motor (起动机)

4. Added Resistance (附加电阻)

5. Ignition coil (点火线圈)

6. Breaker (断电器)
7. Distributor (配电器) } Stacked on same shaft

8. Spark plug (火花塞)

9. Accumulator (蓄电池)

10. Capacitor (电容器)

11. Low voltage coil lead (低压导线)

12. High voltage coil lead (高压导线)

1 Fundamental of Conventional Ignition System

传统点火系统的工作原理

动画见另一份PPT（汽车构造）

Requirements of Spark plug

(1)adequate **mechanical strength**(足够的机械强度)

(2)**sufficient insulation** strength(足够的绝缘强度) can bear high volatage-- 30 kv.

(3)not only high temperature resistant(耐高温), but also can bear the **temperature change**(承受温度剧变).

(4)**corrosion** resisting(耐腐蚀).

(5)the **appropriate electrode gap**(合适的电极间隙) and the **appropriate position**.

Thermal characteristics of spark plug

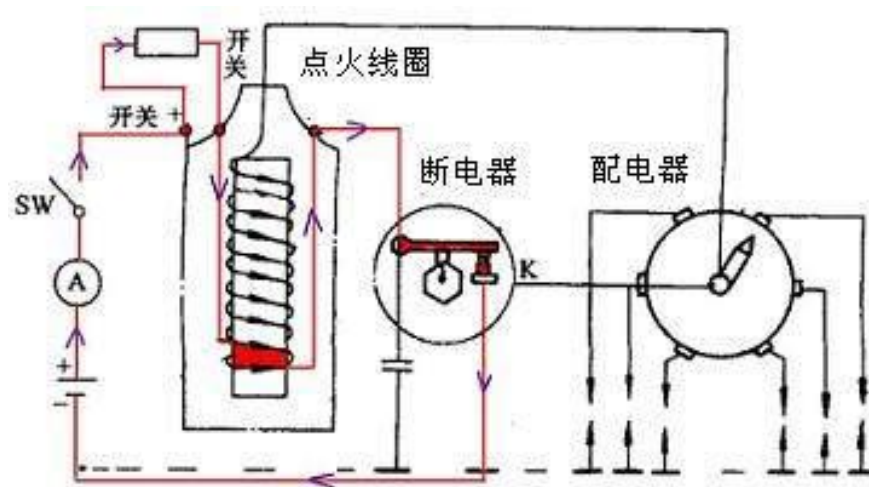
Self-purification temperature(自净温度): can make the fallen droplets on the insulator burnt immediately, not form a carbon deposit(积炭), temperature of insulator skirt (绝缘体裙部)should maintain 500 ~ 600 ° C.

Basic working process

(1) Low voltage circuit 低压回路:

Contact K close: positive electrode → ignition switch → additional resistance → primary coil → moving contact arm of Breaker → contact (capacitor) → distributor block GND → negative electrode.

触点K闭合时（充电）：蓄电池正极 → 点火开关 → 附加电阻 → 初级线圈 → 断电器活动触点臂 → 触点（电容器） → 分电器壳体搭铁 → 蓄电池负极。



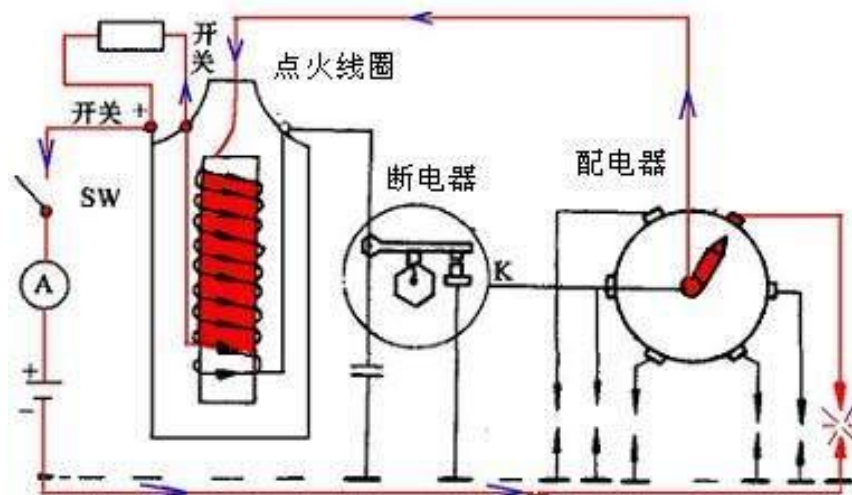
动画见另一份PPT（汽车构造）

Basic working process

(2) High voltage circuit 高压回路:

Contact K open: secondary coil → additional resistance → positive electrode → battery → GND → spark plug side electrode → spark plug center electrode → high voltage wire → distributor block side electrode → ignition divide plug → central contact → central high voltage wire → secondary coil.

触点K断开（放电）：次级线圈 → 附加电阻 → 蓄电池正极 → 蓄电池 → 搭铁 → 火花塞侧电极 → 火花塞中心电极 → 高压线 → 配电器旁电极 → 分火头 → 中心触点 → 中央高压线 → 次级线圈。



动画见另一份PPT（汽车构造）

3) Additional Resistance(附加电阻)

◆ Features: thermal resistance(热敏电阻), its resistance value is direct proportion to temperature(正比).

◆ Function: primary current(初级电流) is automatic adjusted, which can improve the performance of the ignition system.

✓ Low speed: the current value (I) $\uparrow \rightarrow$ Temperature (T) $\uparrow$ additional resistance value(R) $\uparrow \rightarrow$ the primary current(I) $\downarrow$ prevent ignition coil overheating;

✓ High speed: $I \downarrow \rightarrow T \downarrow \rightarrow R \downarrow \rightarrow$ the primary current drops less, ensure ignition system can supply high enough voltage, avoid misfire.

✓ Cold Start: the battery terminal voltage (端电压) $\downarrow$ the primary current $\downarrow$, ignition coil can't supply high enough voltage and energy. Additional Resistance short circuit(短路, bypassed) $\rightarrow$ the primary current $\uparrow \rightarrow$ the secondary voltage and sparking energy $\uparrow$.



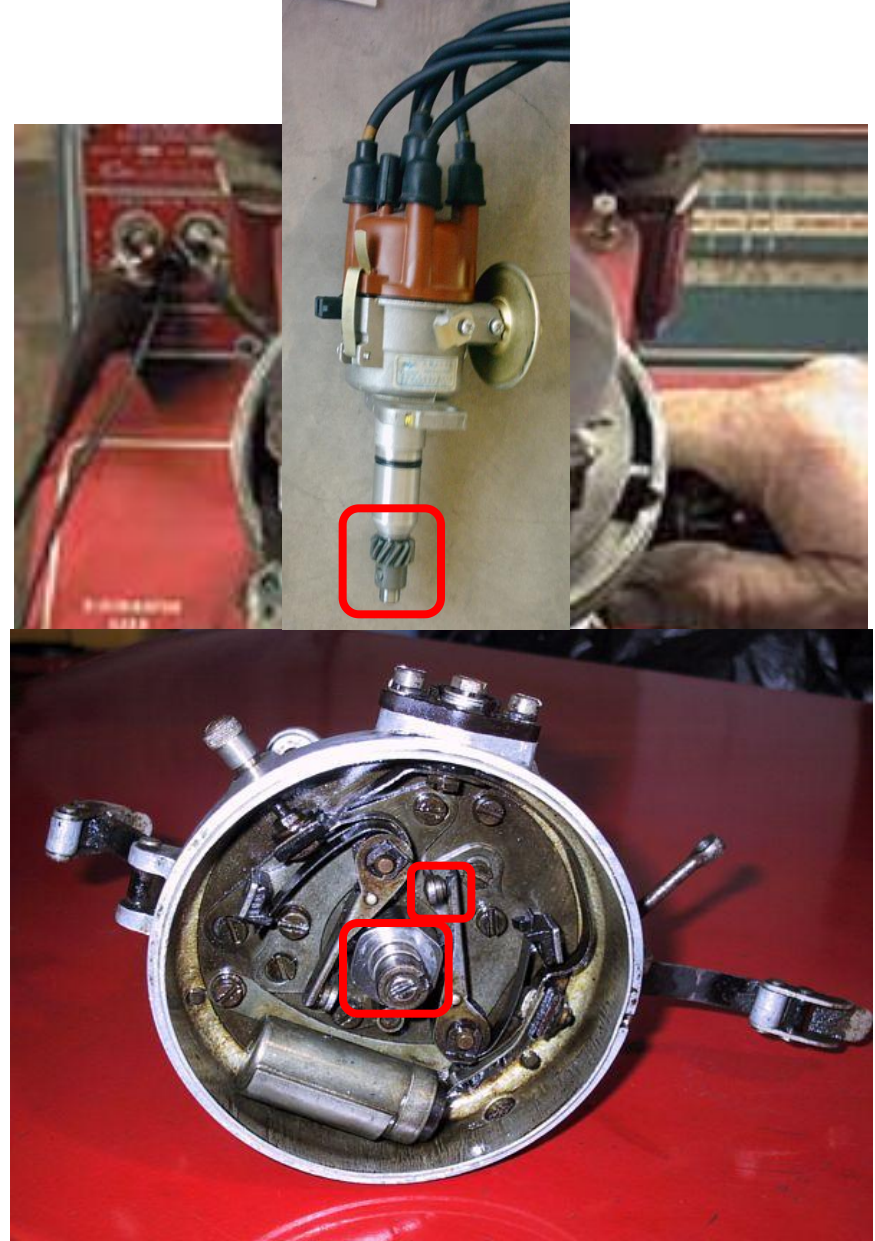
4) Distributor block (分电器)

■ Composition:

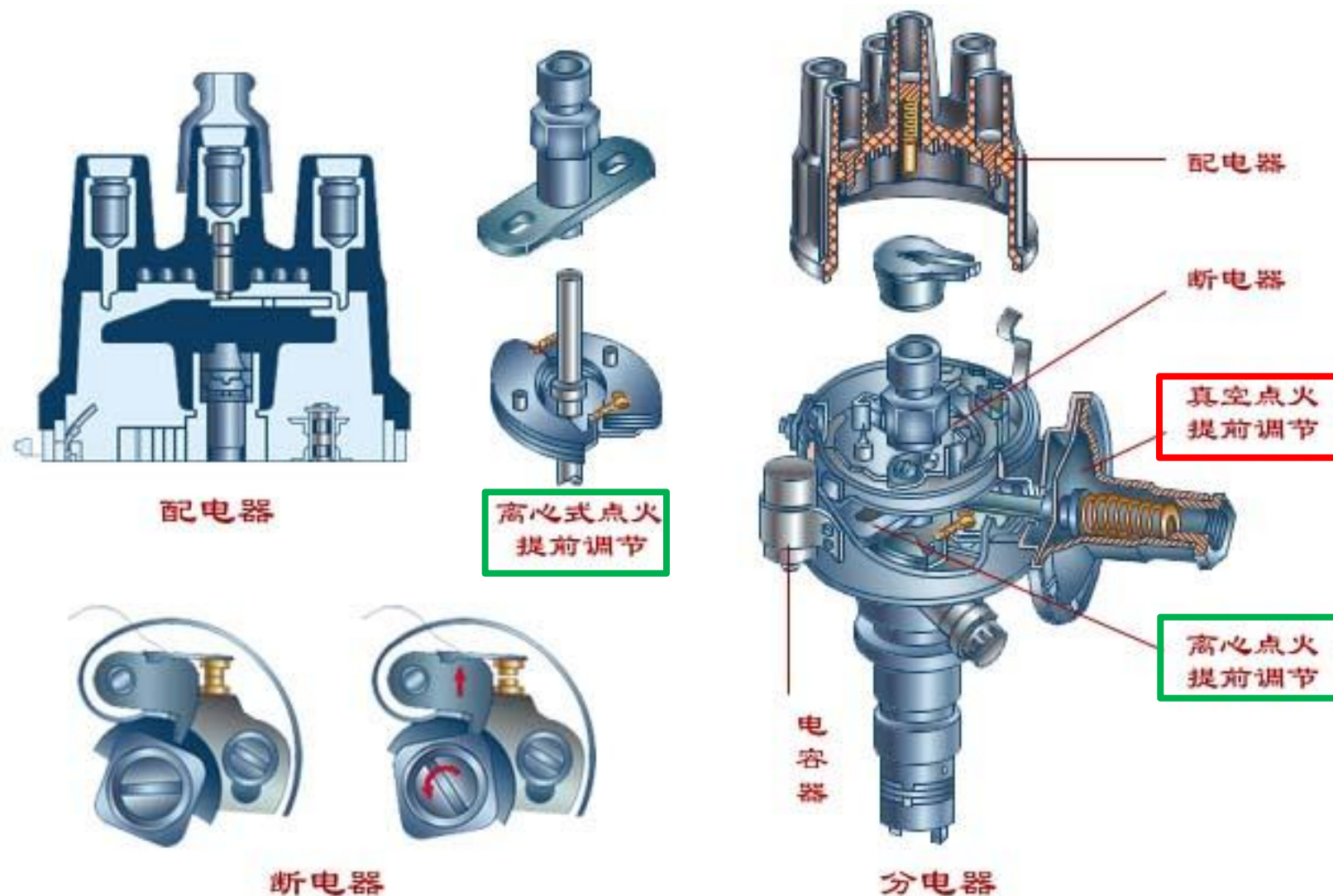
Breaker(断电器), Distributor(配电器), capacitor(电容器) and Ignition advancer control device(点火提前调节装置).

■ Function:

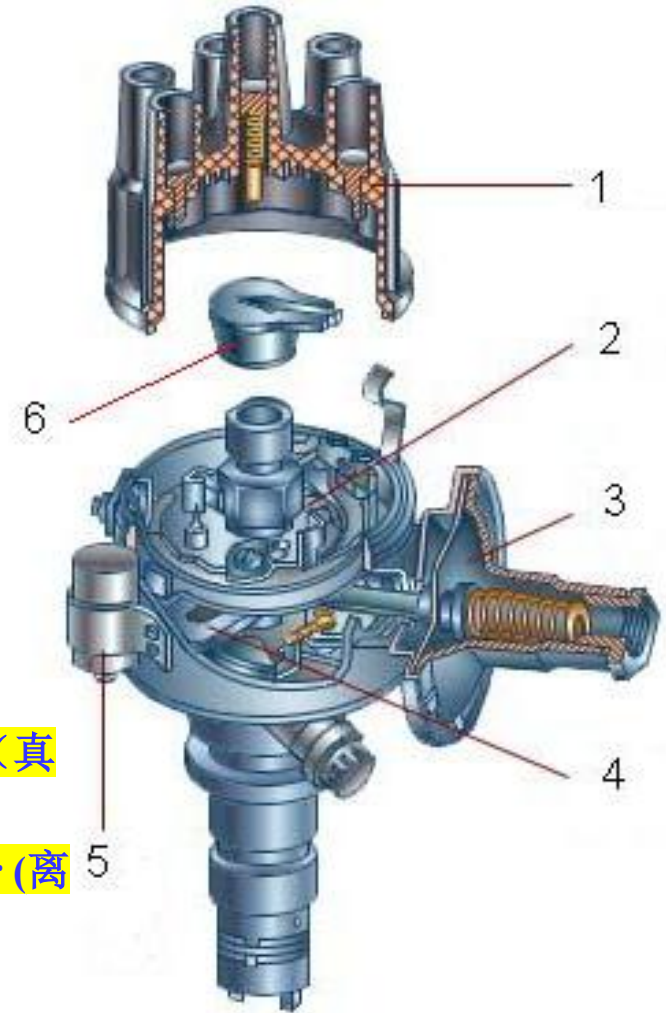
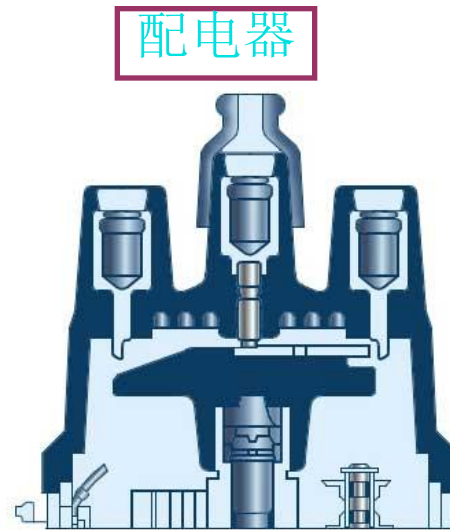
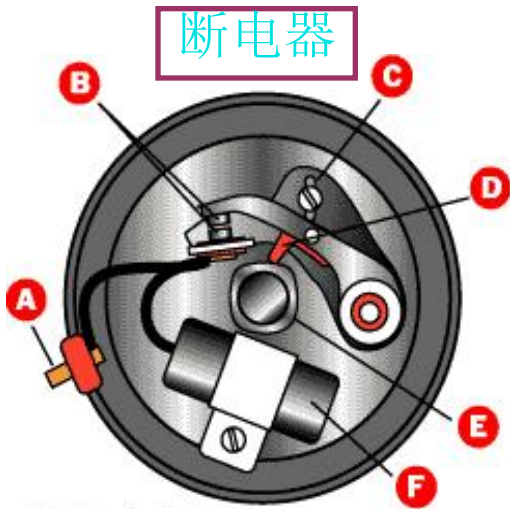
- ① connect and cut off the primary circuit of the Ignition system , make the secondary winding (次级绕组) of the ignition coil produce high voltage;
- ② distribute the high voltage produced by the ignition coil to corresponding cylinder spark plug.
- ③ adjust the ignition timing according to engine speed and load automatically.



Composition of Distributor block



Composition of distributor block



- A. connection to coil (接线柱)
- B. breaker point (断电器触点)
- C. adjusting screw (调整螺钉)
- D. cam follower (凸轮从动件)
- E. distributor cam (分电器凸轮)
- F. capacitor (电容器)

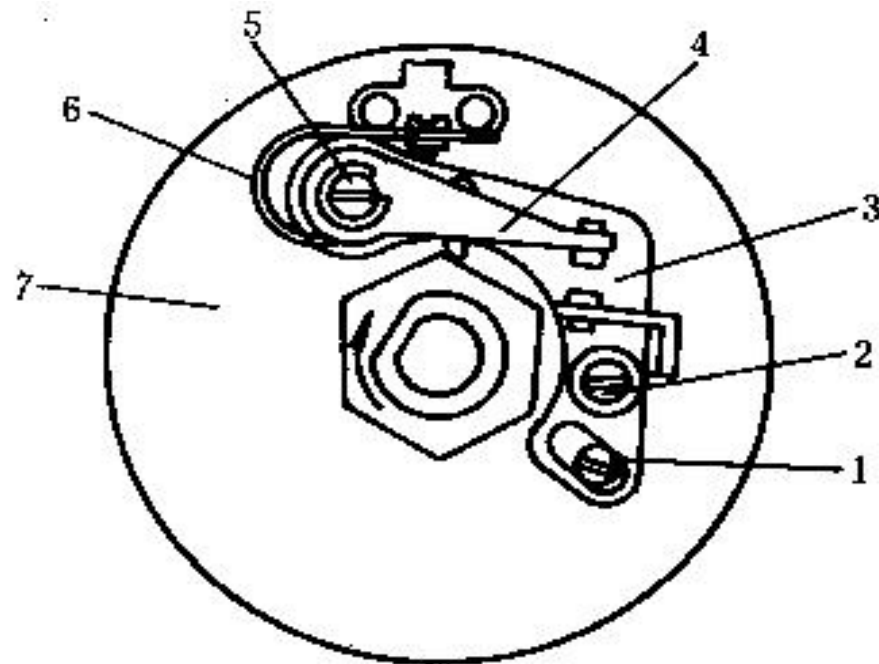
- 1.distributor (配电器)
- 2.breaker (断电器)
- 3.vacuum ignition advancer (真空点火提前装置)
- 4.centrifugal ignition advancer (离心点火提前装置)
- 5.capacitor (电容器)
- 6.Ignition divide plug (分火头)

① Breaker (断路器)

(1) when **contact gap** (触电间隙) is too small, spark appears in the gap easily and makes the **primary circuit power bad**, and even **contact ablation** (触点烧蚀);

(2) when the contact gap is too large, time of contact closure (触点闭合时间) is shortened, the primary current decreases, and secondary voltage(次级电压) is reduced, misfire easily at high speed.

(3) adjustment: rotate eccentric screw 1(偏心调节螺钉1) to adjust contact clearance.



- 1.adjusting screw(调节螺钉)
- 2.fastening screw(固定螺钉)
- 3.layer board (托板)
- 4.moving contact arm(活动触点臂)
- 5.pin(销钉)
- 6.leaf spring(片簧)。

☆ CAM edge number 凸轮棱数 = number of cylinders 气缸数

☆ camshaft rotate speed 凸轮轴转速 = rotate speed of camshaft 配气凸轮轴转速

② Distributor (配电器)

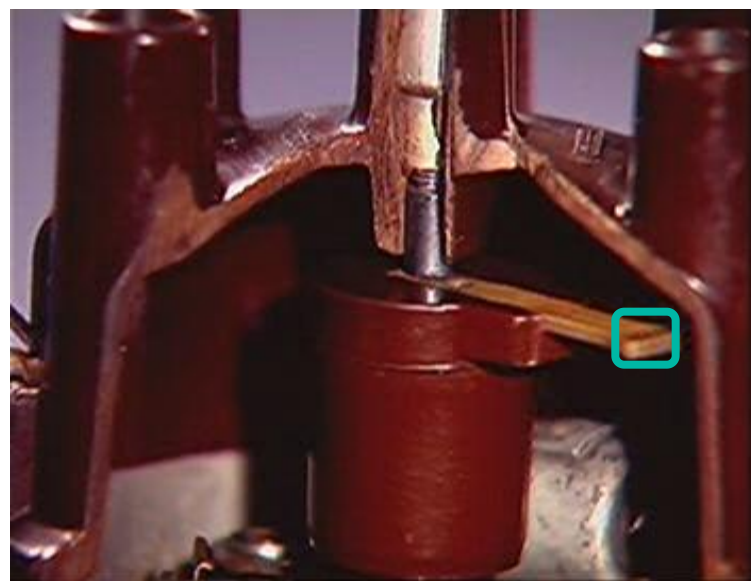
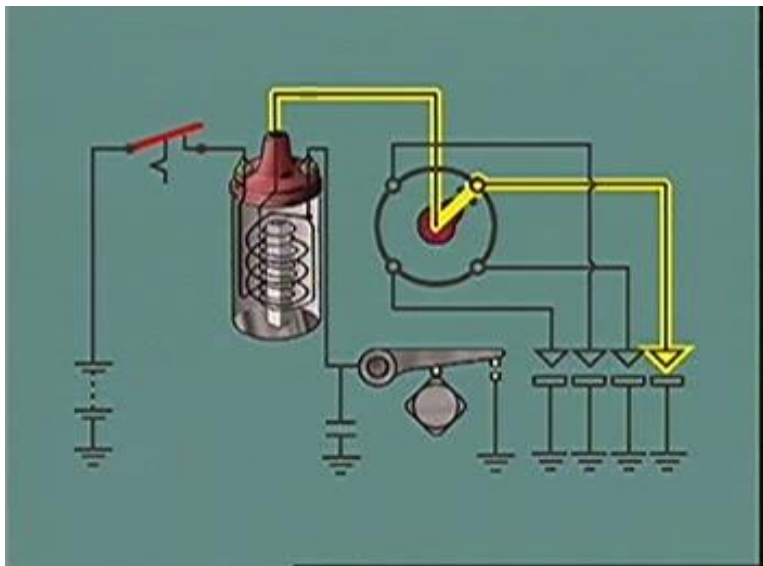
◆ Composition:

Distributor block cap, ignition **divide plug**.

分电器盖、分火头.

◆ Function:

Distribute the high voltage to spark plug according to **ignition sequence** (点火顺序).

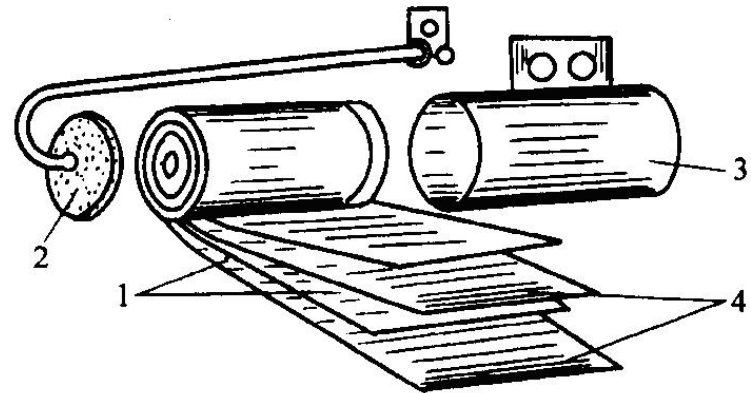


③ Capacitor (电容器)

◆ Function:

(parallel with Breaker与断路器并联)

When contact is off, **reduce electric spark between the contacts**, avoid contact being ablated(触点烧蚀). At the same time, absorb/draw self induction electromotive force(自感电动势) of primary winding, cut off the primary current quickly and increase secondary voltage.



1-metal foil(金属箔) 2-cover plate(盖板) 3-Enclosure(外壳)
4-insulation paper(绝缘纸)

④ Ignition advancer device （点火提前装置）

➤ Function

Realize ignition advanced, make contact apart in the compression stroke before TDC.

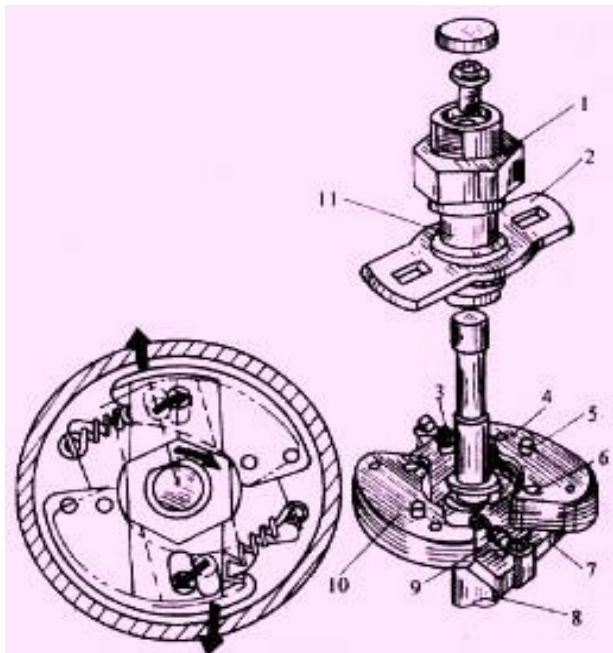
➤ Methods

1. Contact motionless, make CAM (nut) turn an angle relative to shaft;
触点不动，使凸轮相对相对轴向转一个角度；
2. CAM motionless, make contact turn an angle relative to CAM.
凸轮不动，使触点相对凸轮转一个角度。

➤ Device

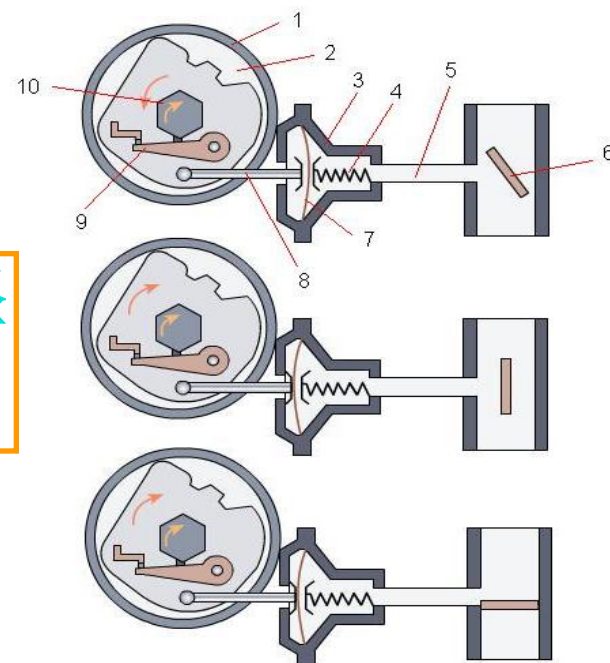
1. Centrifugal ignition advancer – engine speed
离心式点火提前调节装置 – 响应转速变化
2. Vacuum ignition advancer – engine load
真空式点火提前调节装置 – 响应负荷变化
3. Octane number corrector – fuel difference
辛烷值校正器 – 针对燃料区别

离心点火 提前调节 装置



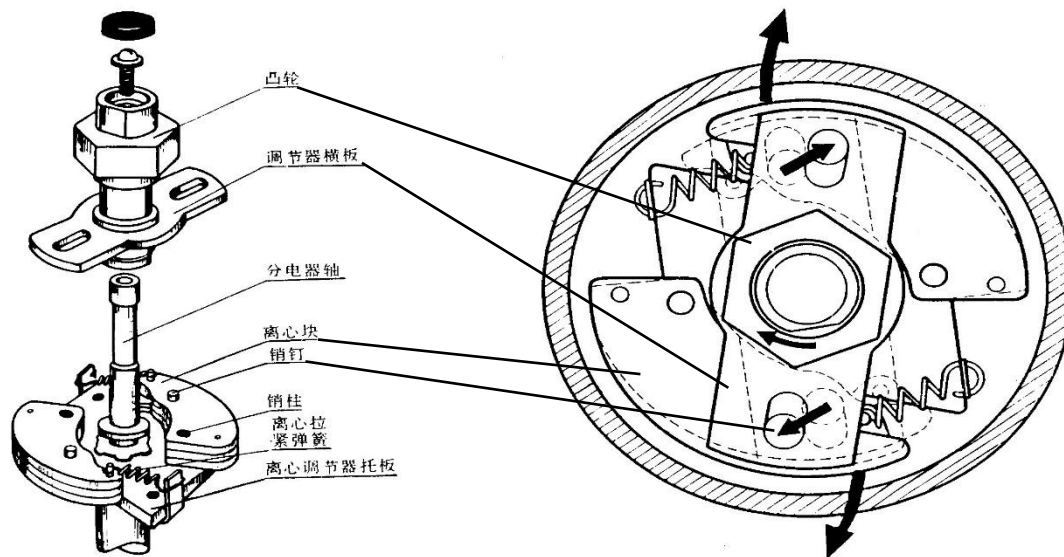
- | | |
|--------------------|----------------------------|
| 1.cam (凸轮) | 6.axle pin (轴销) |
| 2.touch panel (拨板) | 7.bearing plate (托板) |
| 3.9.spring (弹簧) | 8.distributor shaft (分电器轴) |
| 4.10.block (重块) | 11.spindle sleeve (轴套) |
| 5.pin bolt (销钉) | |

真空点火 提前调节 装置

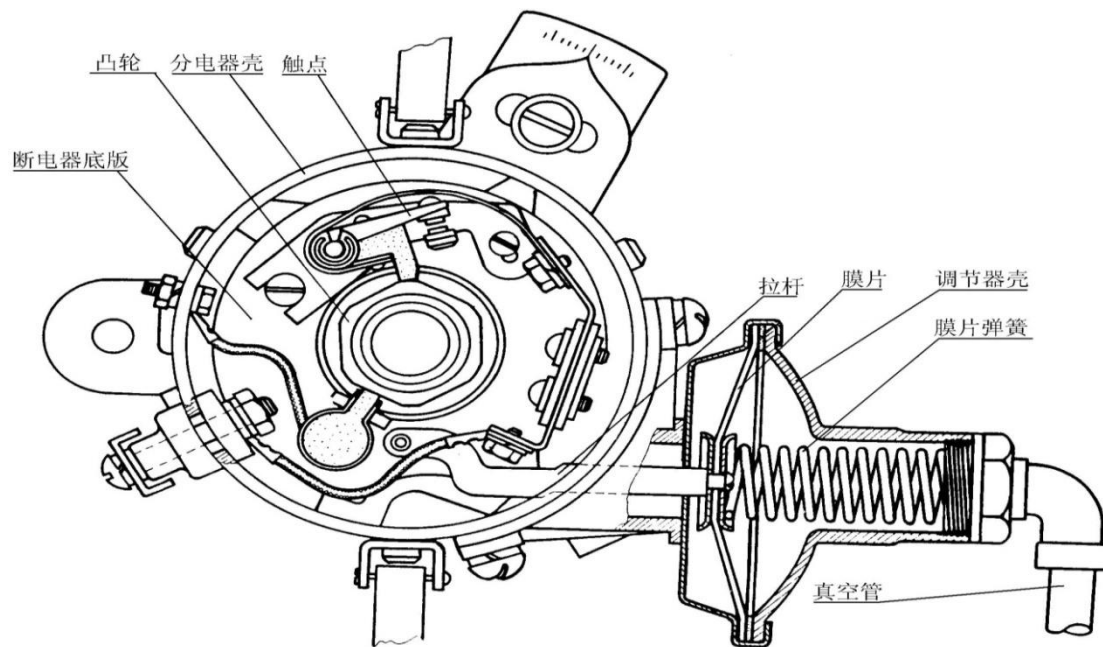


- | | |
|---|----------------------|
| 1.distributor shell (分电器壳) | 6.air throttle (节气门) |
| 2.breaker base (断电器底板) | 7.diaphragm (膜片) |
| 3.vacuum ignition advancer shell (真空点火提前装置外壳) | 8.nutted rod (拉杆) |
| 4.spring (弹簧) | 9.contact (触点) |
| 5.connecting pipe (连接管) | 10.cam (凸轮) |

离心点火 提前调节 装置

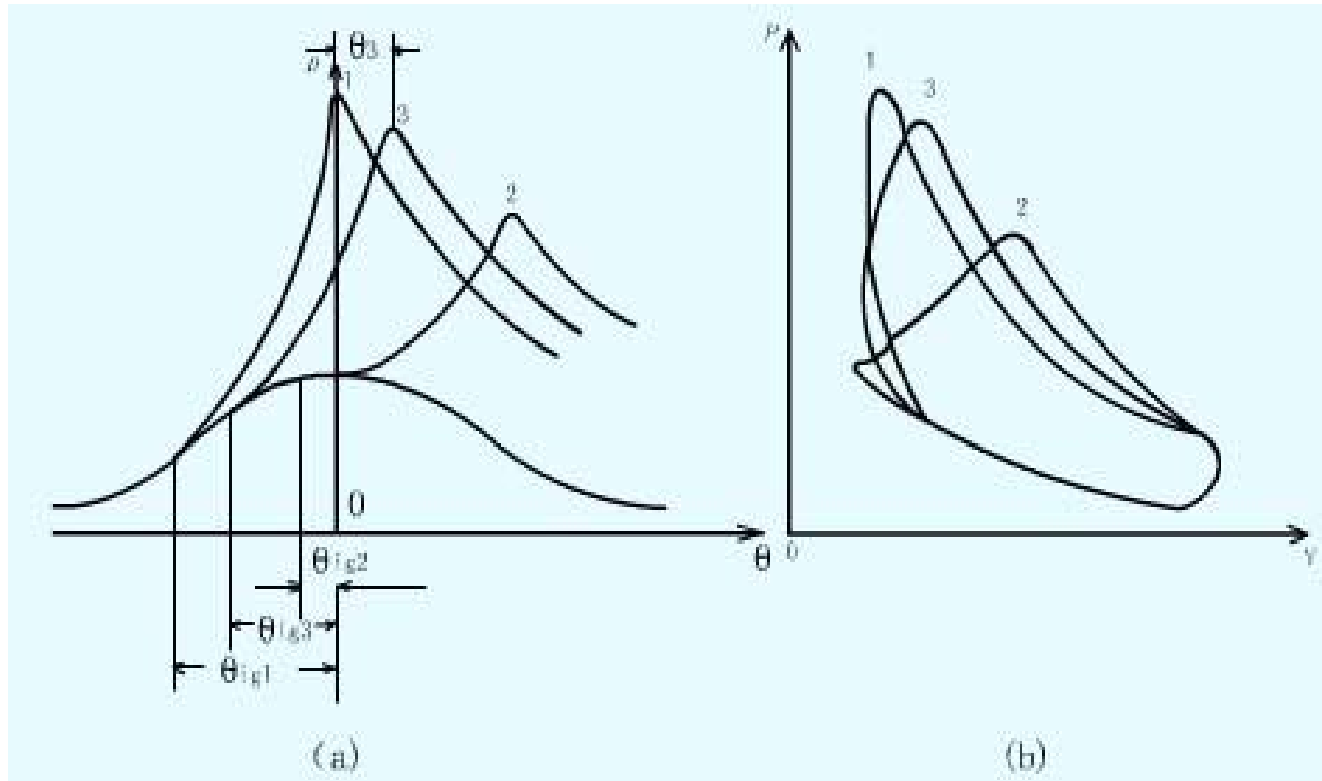


真空点火 提前调节 装置



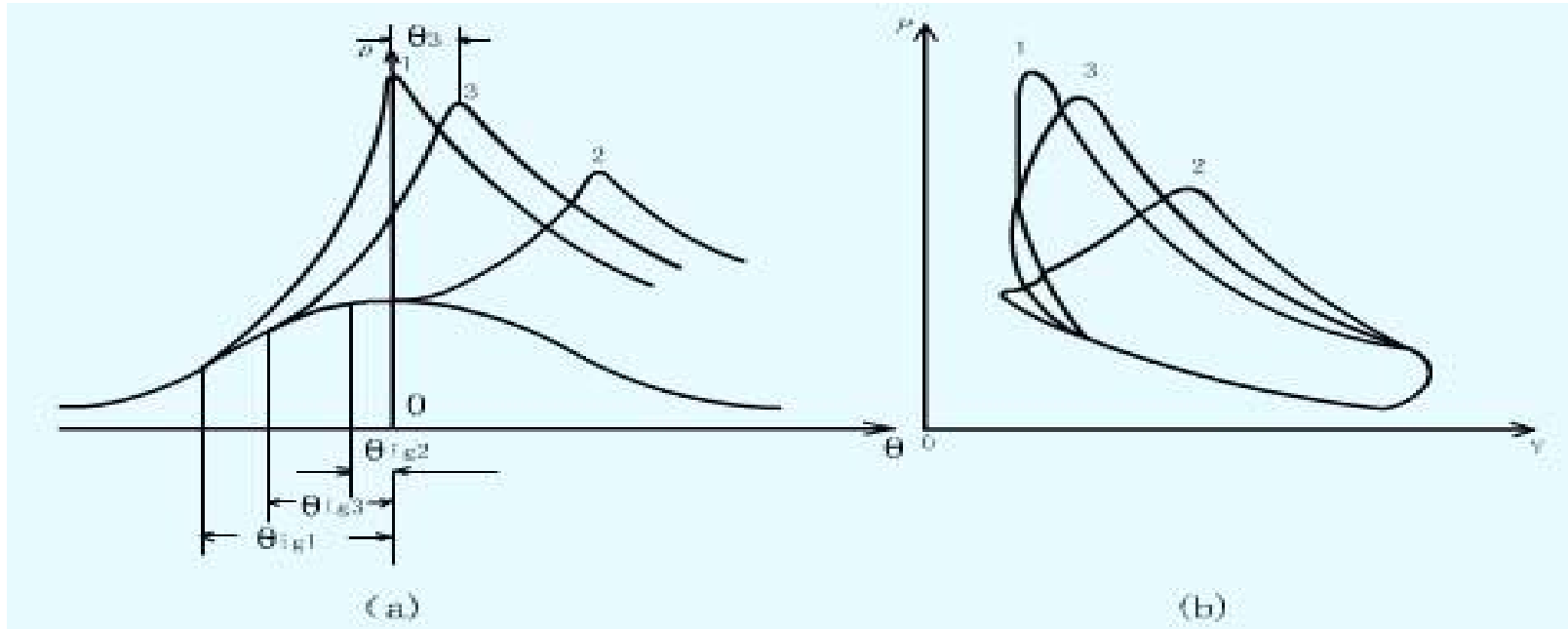
Ignition advanced Angle (点火提前角)

Definition: Crank Angle from the spark to TDC of compression stroke.



- 1、 Ignition timing is too advance(过早) ;
- 2、 Ignition timing is too late (过晚) ;
- 3、 The best Ignition timing(最佳) .

Ignition advanced Angle (点火提前角)



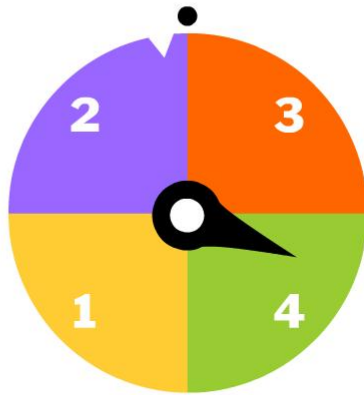
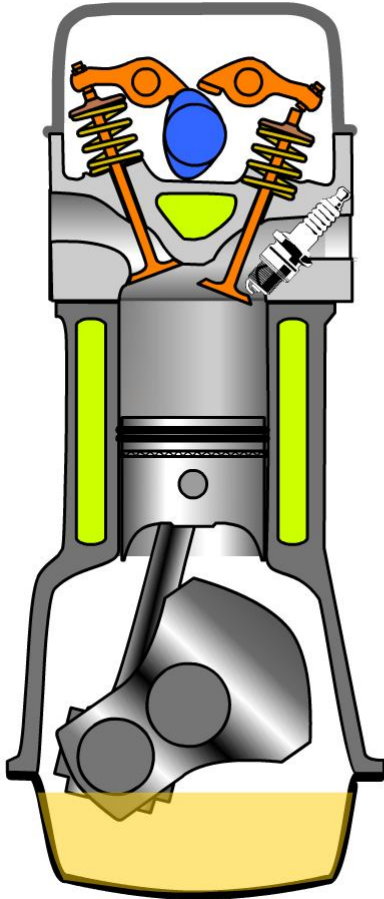
(1) too advance(过早): most of the mixture is burned in the compression stroke, compression work $\uparrow$, the engine easily to be overheat, $P_e\downarrow$, work coarse detonation degree(工作粗暴程度) $\uparrow$. engine knock(爆震) $\uparrow$.

(2) too late (过晚): the combustion extends to the expansion process, $P_{\max}$ 、 $T_{\max}\downarrow$, heat transfer loss $\uparrow$, exhaust temperature $\uparrow$, $P_e\downarrow$, $b_e\uparrow$.

(3) the optimal value(最佳): $P_{\max}$ appears in the **right Angle ATDC10~12°**, indicator diagram area (示功图面积) is the largest, W 、 P_e 、 b_e are the best.

Ignition advanced Angle (点火提前角)

© 2001 HowStuffWorks, Inc.



Factors:

◆ Engine load

When Engine speed is constant, load $\uparrow$
mixture into the cylinder $\uparrow$ Ignition
advanced Angle $\downarrow$

◆ Engine speed

When throttle angle is a constant, Engine
speed $\uparrow$
crank angle $\uparrow$ Ignition advanced Angle $\uparrow$

◆ Fuel anti-knock rate (抗爆性)

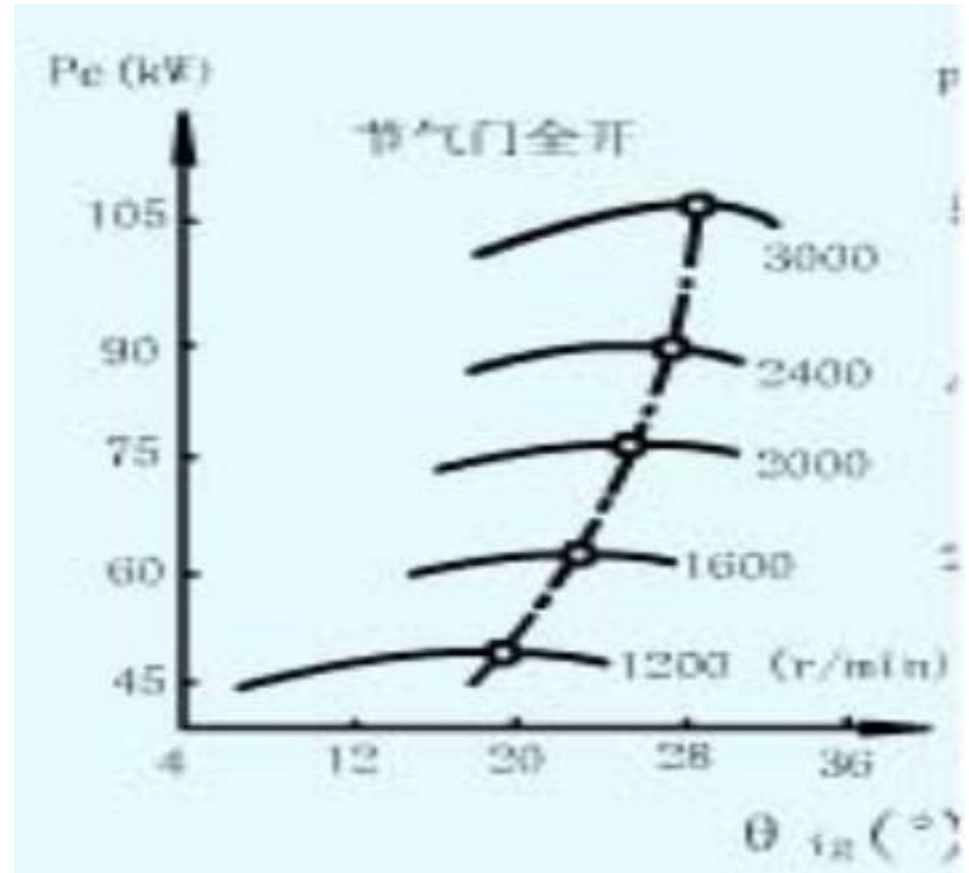
Octane number(辛烷值) $\uparrow$ anti-knock ability $\uparrow$
Ignition advanced Angle $\uparrow$

① Engine speed

When load is a constant,

$n \uparrow$, turbulent flow $\uparrow$, the flame propagation speed (火焰传播速度) is proportional to speed, so the combustion process is shortened, the cycle time $\downarrow$, crankshaft Angle of combustion process increased----**increase ignition advanced Angle**, corresponding to **centrifugal ignition advancer** (离心调节点火提前器).

Engine Speed increases, the flame speed also increases, the deflagration (爆燃) tends to decrease.

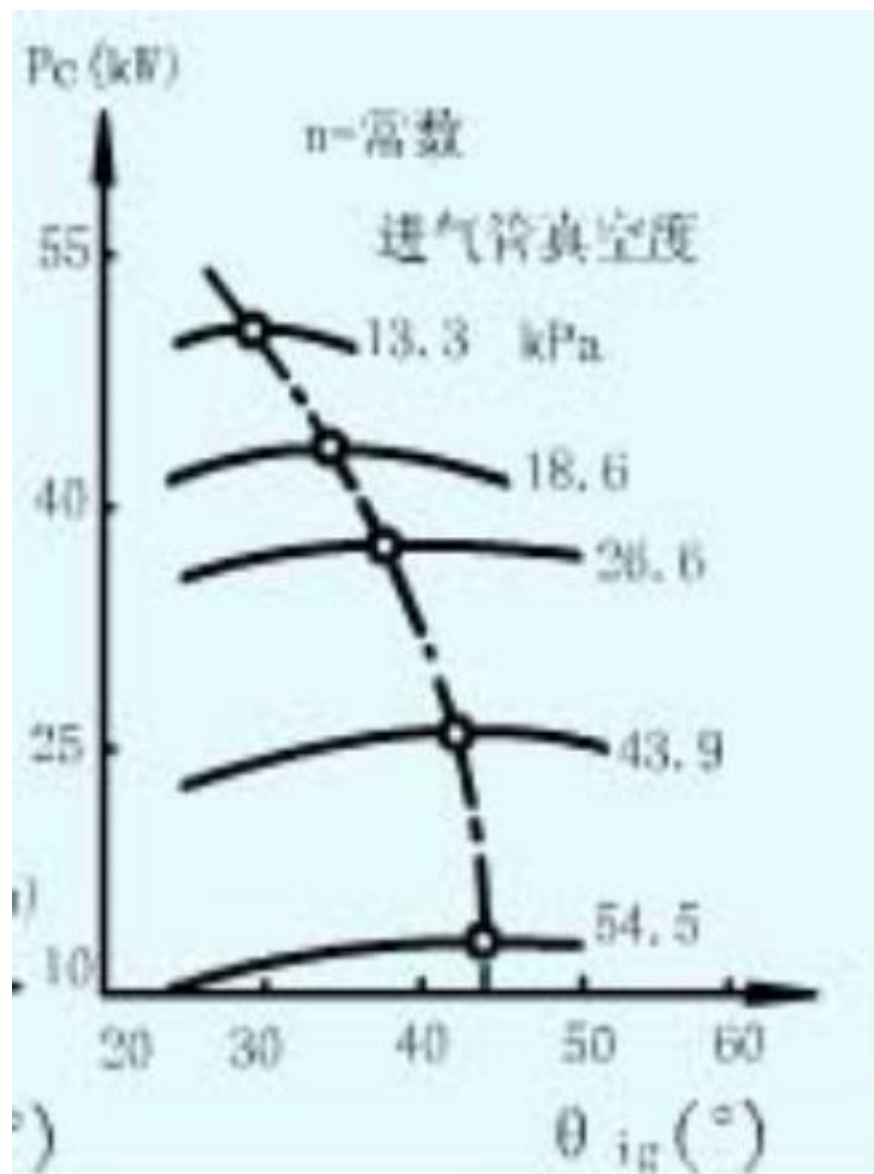


② Engine load(负荷)

When engine speed is a constant, load↓, fresh charge into the cylinder↓, the residual gas coefficient(残余废气系数) ↑, --- --increase ignition advanced Angle.

Corresponding to **vacuum ignition advancer**;

(真空式点火提前调节在装置)



§ 10.3 Electronic Ignition system(电子点火系统)

Advantages :

- 1) reduce the **contact spark(触点火花)**, avoid contact being **ablated(烧蚀)**, some can cancel contact, improve the ignition performance.
- 2) can not be restricted by contact, ↑primary current, ↑ the secondary voltage, ↑ engine ignition performance at high speed.
- 3) due to the improvement of secondary voltage and ignition energy, make its insensitivity to spark plug **carbon deposit(火花塞积炭)**, and can increase the spark plug electrode gap.
- 4) **reduce the disturbance to the radio greatly.(减轻了对无线电的干扰).**
- 5) simple structure, light weight, small volume.

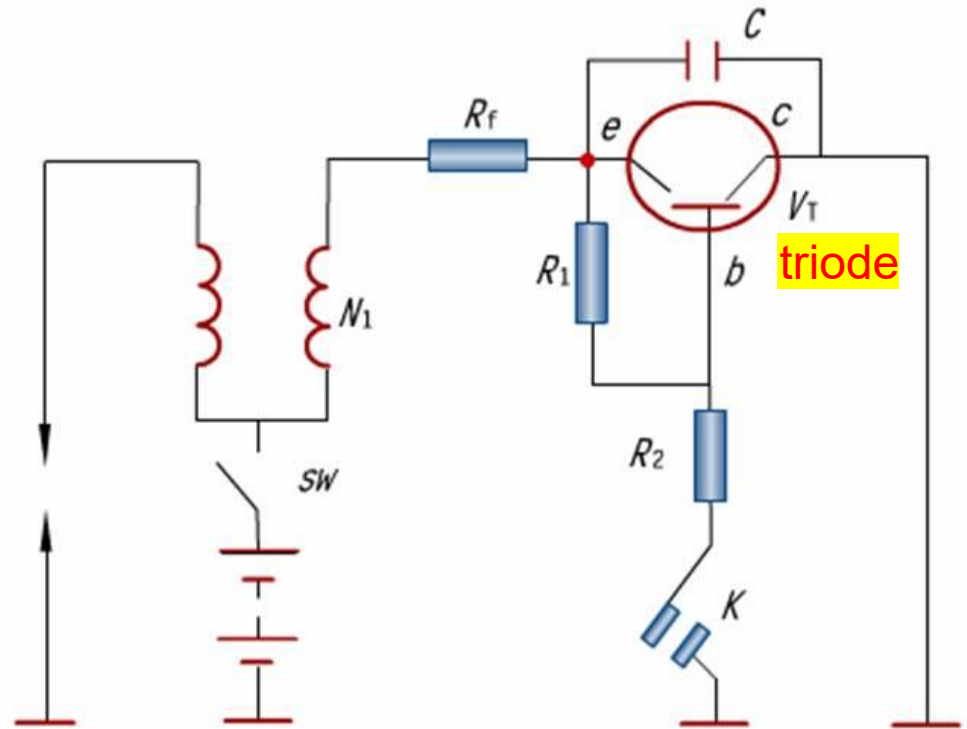
1 Electronic Ignition System with Contact(有触点电子点火系统)

◆ Principle

by reducing contact current(触点电流) and decreasing contact spark(触点火花), improving the ignition performance.

the current through the contact (触点电流) is $1/5 \sim 1/10$ of primary current (一次电流).

can not eliminate other defects of contact, it's rarely used.



有触点式电子点火系统的电路原理图

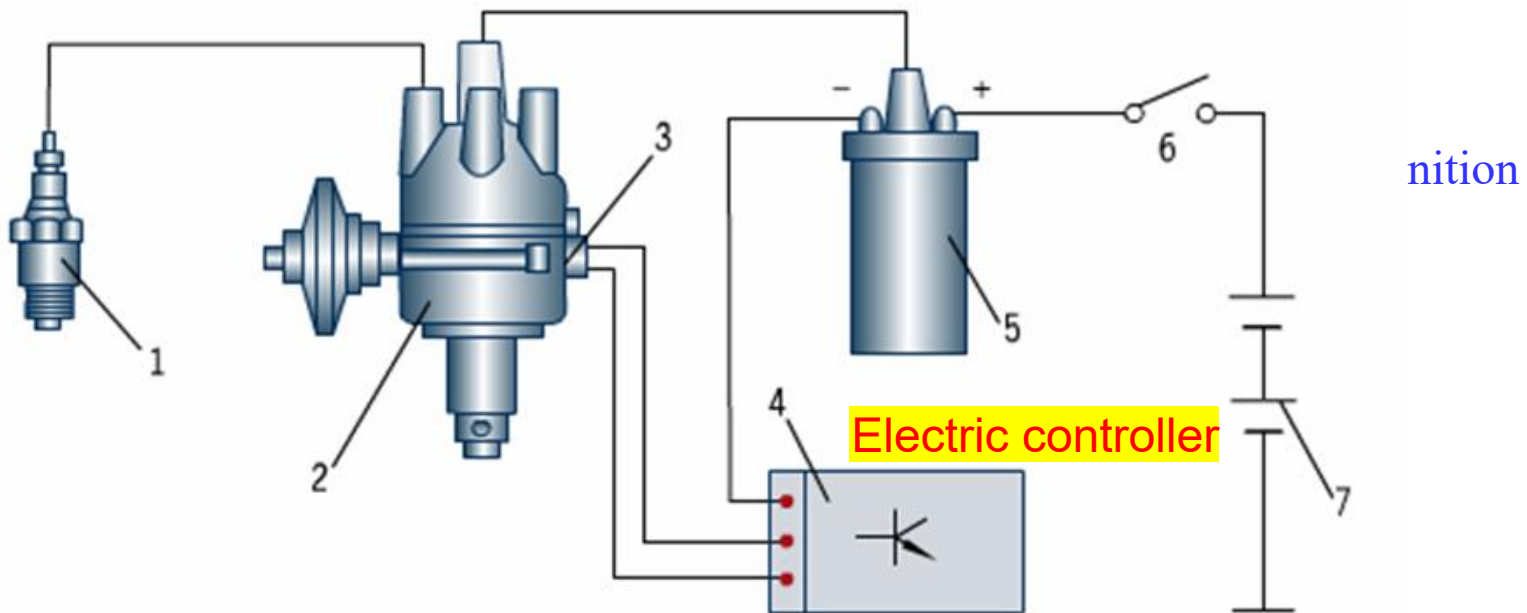
2 Electronic Ignition System without Contact(无触点电子点火系统)

◆ Principle

Ignition signal produced by sensors , overcome all the shortcomings concerned with the contact, use

◆ Components

Ignition signal sensor, distributor, coil, distributor



无触点电子点火系统的组成

1-火花塞； 2-分电器； 3-点火信号发生器； 4-点火控制器；
5-点火线圈； 6-点火开关； 7-电源

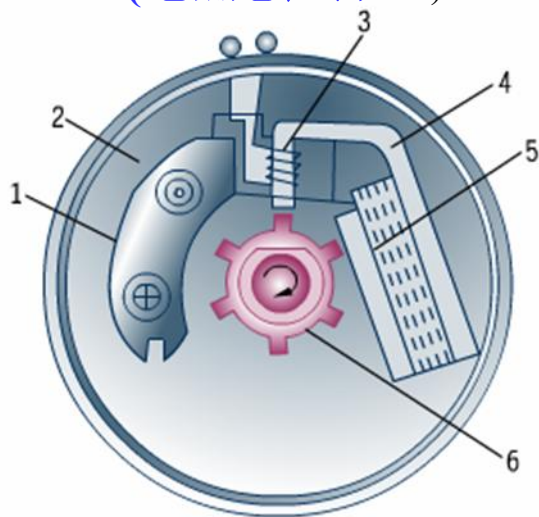
(1) Sensors

Types:

Magnetic induction, hall effect, photoelectric effect

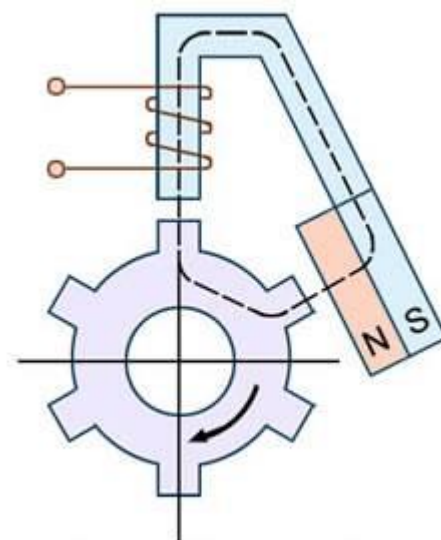
磁感应式、霍尔效应式和光电效应式。

◆Magnetic induction (磁感应式) is based on Principle of electromagnetic induction(电磁感应原理)



磁脉冲式点火信号发生器

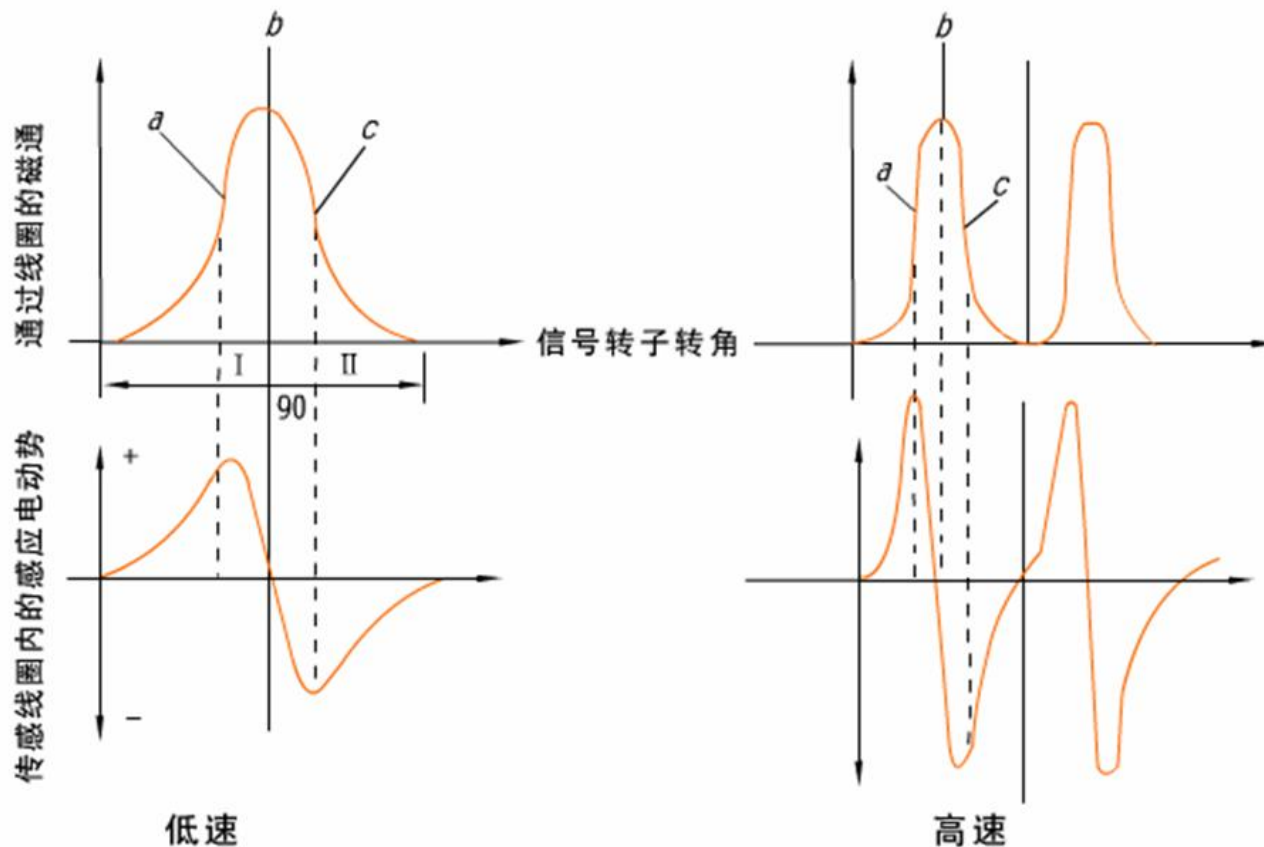
1-底板； 2-活动底板； 3-传感线圈；
4-铁心； 5-永久磁铁； 6-信号转子



磁脉冲式点火信号发生器的工作原理图

①Magnetic induction (磁感应式)

Characteristic: $n \uparrow \rightarrow V \uparrow$ (0.5 ~ 100 V)



低速

高速

不同转速时传感线圈内磁通及磁感应电动势的变化情况

② Photoelectric effect (光电效应式)

Components:

the cover disc (signal of the rotor), blanking disc axle, light source, optical receiver (photosensitive components).

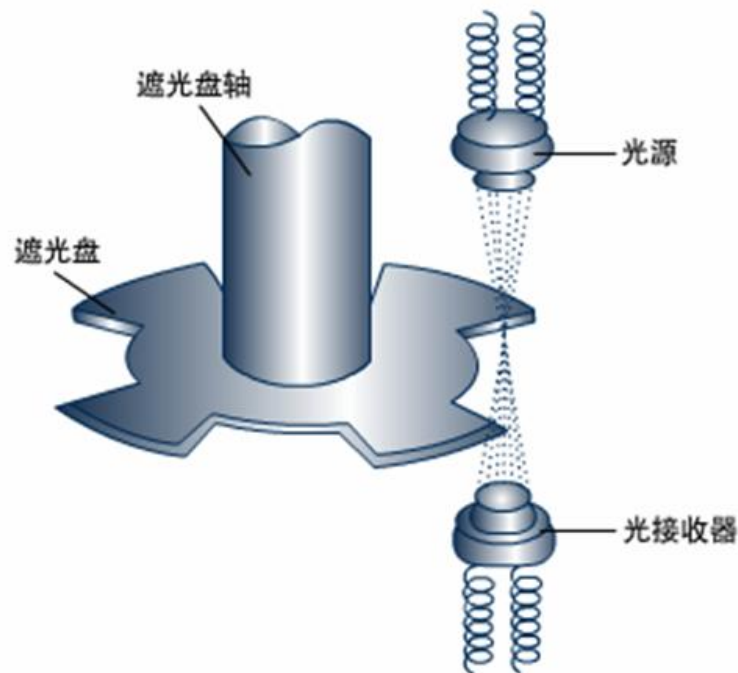
主要由遮光盘(信号转子)、遮光盘轴、光源、光接收器(光敏元件)等组成。

◆ Light source: incandescent lamp、light-emitting diodes (leds).

光源：白炽灯，也可用发光二极管

◆ Optical receiver: photosensitive diode, or light activated triode.

光接收器：光敏二极管，或光敏三极管



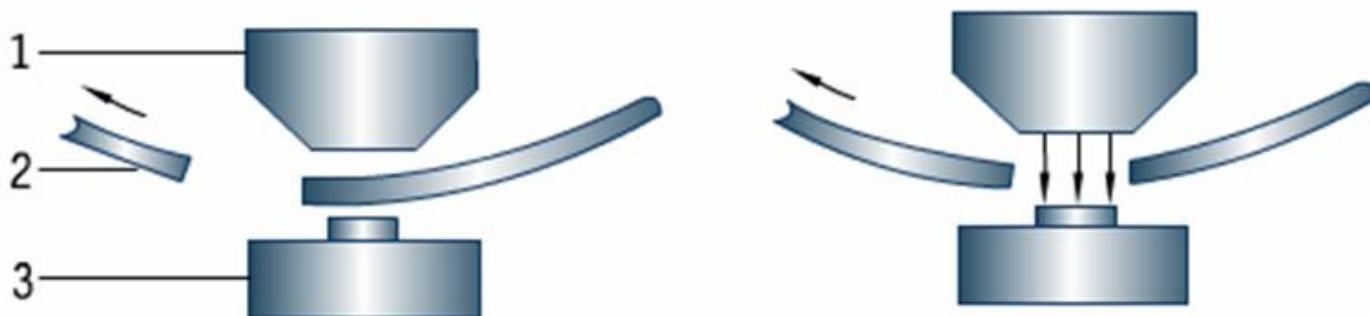
光电效应式点火信号发生器

③ Hall effect(霍尔效应)

Composition:

Installed inside a distributor. consist of the trigger, a permanent magnet and rotor with gap driven by the distributor shaft.

安装在分电器内。由霍尔触发器、永久磁铁和由分电器轴驱动的带缺口的转子（作用等同于断电凸轮）组成。



转子叶片处于永久磁铁和霍尔触发器之间

转子缺口处于永久磁铁和霍尔触发器之间

霍尔效应式点火信号发生器工作示意图

1-永久磁铁； 2-带缺口的转子； 3-霍尔触发器

(2) controller(点火控制器)

signal transformation , amplification, transformed into ignition control signal.
点火信号整形、放大, 转变为点火控制信号

(3) distributor block

Cancel the breaker(contact and CAM)断电器(触点和凸轮) and capacitor(电容器), increased the ignition signal generator(点火信号发生器).

(4) ignition coil

Ignition coil inductance (电感) and resistance (电阻) are smaller than conventional ignition system. Primary current can be much bigger.

(5) spark plug

Ignition energy increased, the spark plug electrode gap is bigger than conventional ignition system

(6) high voltage wire (高压线)

to reduce radio interference(无线电干扰), using high damping line with certain resistance(采用有一定电阻的高压阻尼线).

§ 10.4 Microcomputer Controlled Ignition system

微机控制点火系统

Advantages:

1. Ignition advanced angle is more precise than convential ignition system.
2. Ignition advanced angle is around the best ignition timing due to various of sensors in different load and speed.
3. There are no mechanical wear(机械磨损) and aging（老化） problems.

Types:

with distributor, without distributor

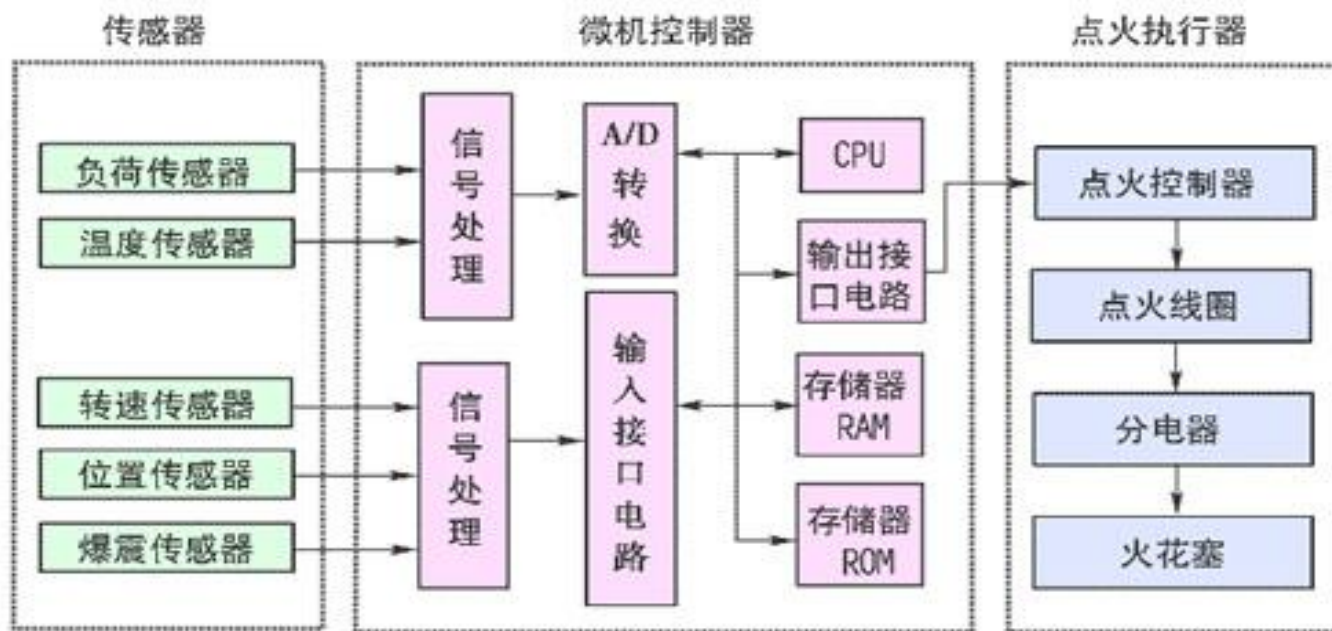
1 Microcomputer Controlled Ignition system with distributor 有分电器的微机控制点火系统

Composition:

Sensors, micro-computer controller, ignition actuators(点火执行器).

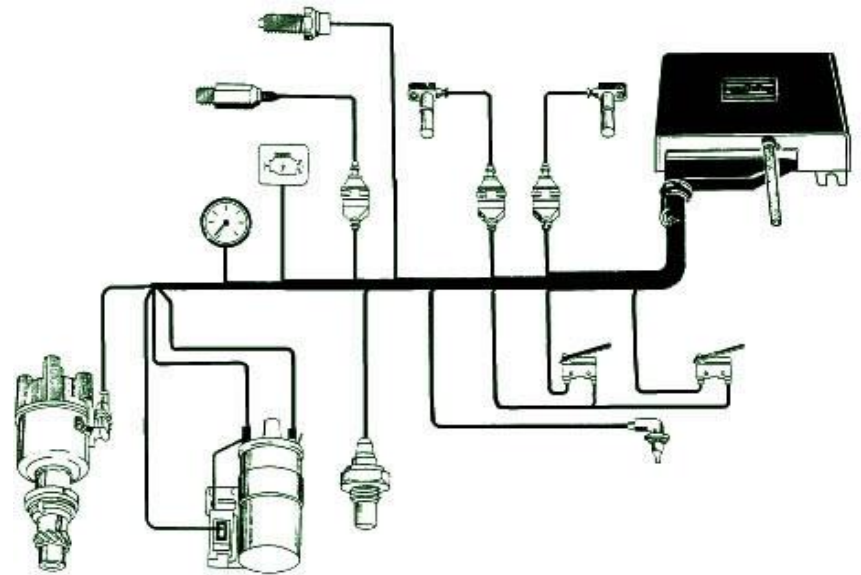
Function:

Ignition advanced Angle, polarization time(通电时间) and explosion control (爆燃控制).



Fundamental(工作原理):

- switch on **ignition switch**;
- start the engine;
- Test conditions and working conditions by sensors;
- ECU calculate ignition Angle and the conduction of time(点火角度和导通时间);
- connect primary curcuit(接通一次电路);
- disconnect primary curcuit;
- secondary curcuit generate high voltage;
- distributor send it to spark plug;
- one cycle;
- **Turn off ignition switch**, Stop engine.



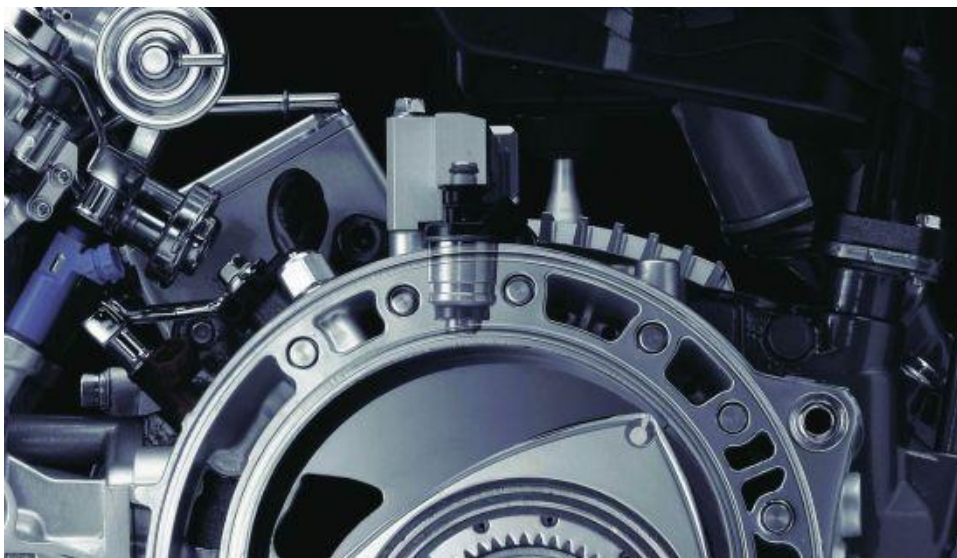
奥迪200微机控制点火系统结构示意图

New Ignition System(新型点火系统)

Laser-induced Spark Ignition System (激光点火系统)

Principle:

Laser ignition process is formed by thermal effect(热作用)、photochemical excitation(光化激发)、resonant decomposition (谐振分解)、non resonant decomposition (非谐振分解) .



Mazda —16X Renesis

◆ Thermal effect(热作用)

When the laser beam irradiate(照射) to the mixture,the translational energy (平动能级)and rotational level (转动能级)or vibration energy (振动能级)of molecule are emitted(激发) → molecules absorb laser energy from transitions between energy levels (能级跃迁), →The laser energy can be converted into heat energy, →the temperature of the mixture is raised to ignite mixture.

◆ Photochemical excitation(光化激发)

Absorb the specific frequency (特定频率) laser photons (光子)and be dissociation(发生离解)→ highly reactive fast particles(高活性快速粒子)→ chemical reaction→ to ignite mixture.

◆ Resonant decomposition(谐振分解)

Photon energy make gas molecules have decomposition(光子的能量促使气体分子分解), →Cause a electric breakdown (电击穿)of the mixture,→plasma formation(等离子区), → produce spark in the laser focal area(激光聚焦区).

◆ Non-resonant decomposition(非谐振分解)

Laser beam focus on one point(聚焦于一点), →creates a strong electric field (电场) around the focus, under the effect of electric field, the gas molecules are broken down →form electric spark ignition.

Types:

Laser-based spark ignition (激光引燃电火花点火)

Conical cavity laser ignition(圆锥形空腔激光点火)

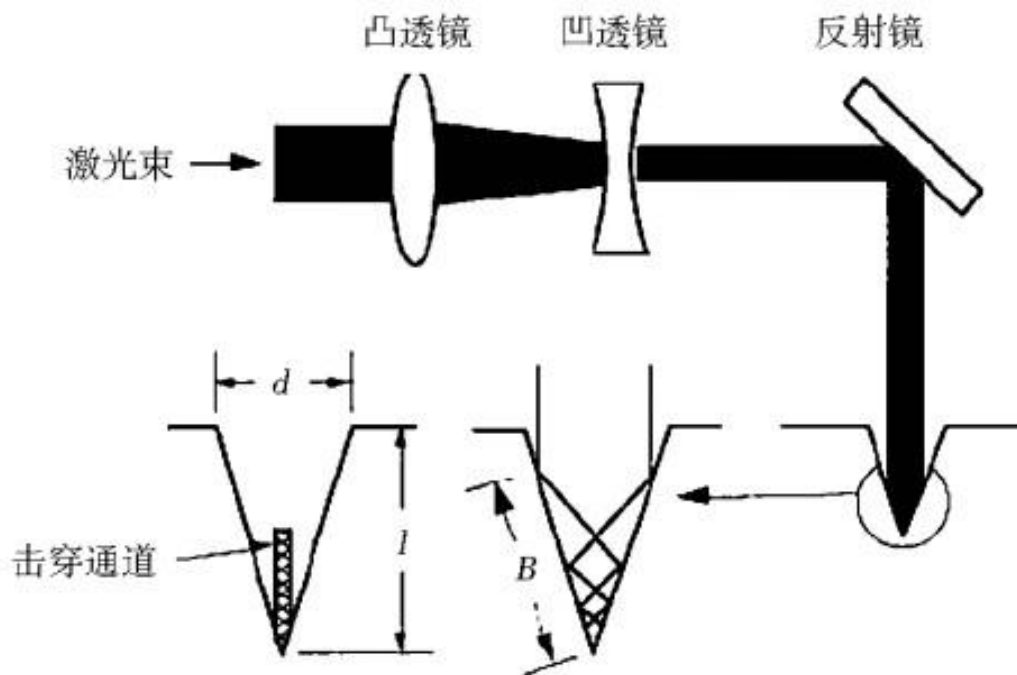


图 1 圆锥形空腔激光点火示意

§ 10.5 Automotive Power Supply (汽车电源)

Composition:

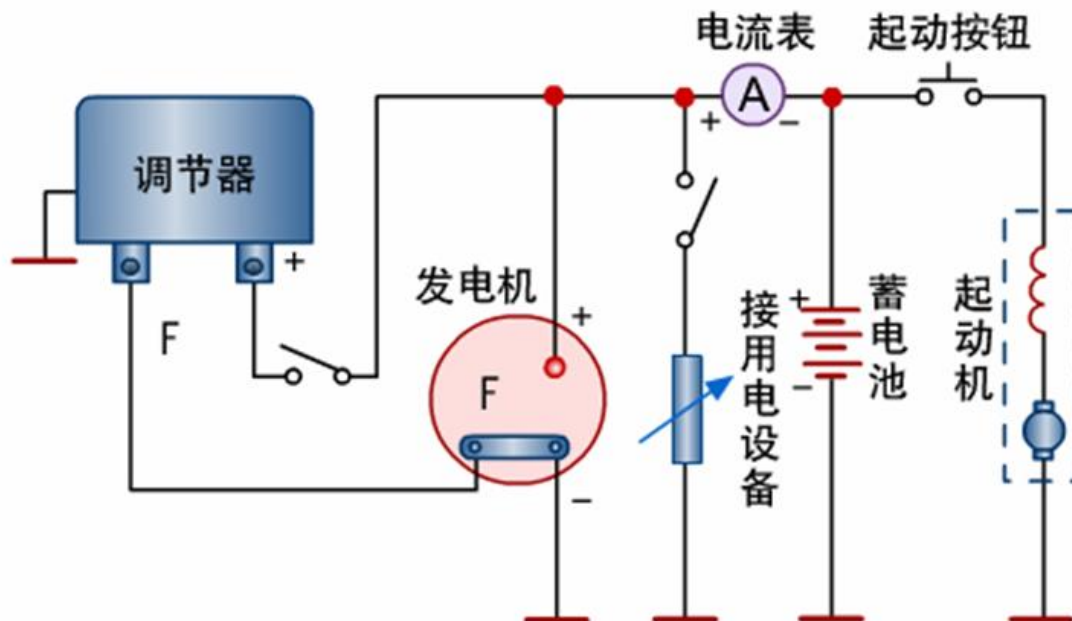
Battery(蓄电池)、Electric Generator(发电机)、voltage regulator(调节器).

3 states:

Start and low speed;

Normal operation;

Big power consumption.

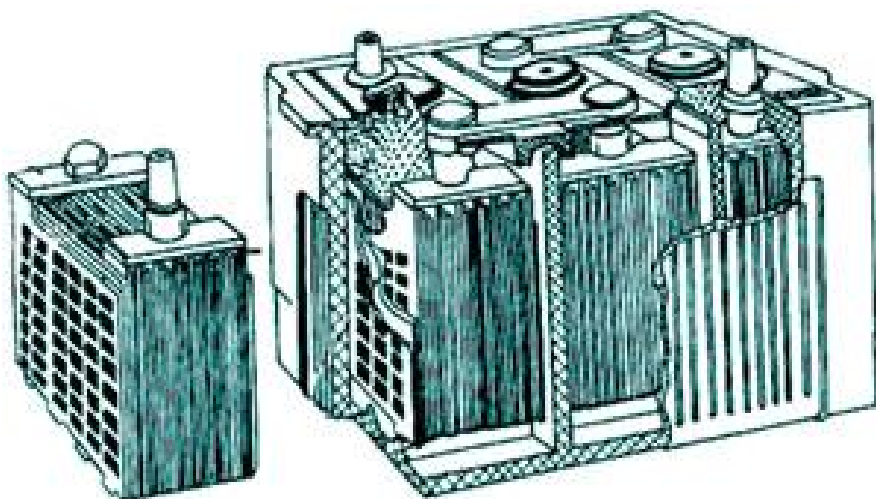


汽车电源电路

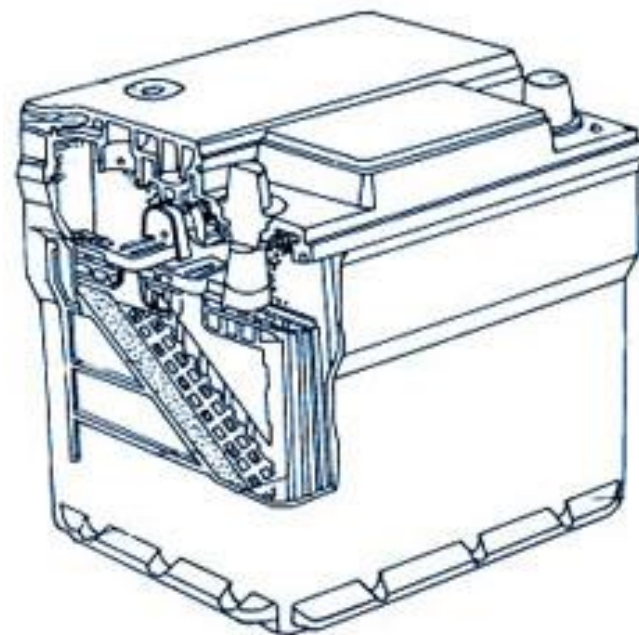
1 Battery(蓄电池)

◆Types: Alkaline and acidic Battery(碱性和酸性蓄电池).

◆Function: Generate power; Store electricity(储存电能);Stabilize the grid voltage(稳定电网电压).



蓄电池



免维护蓄电池的结构图

① Structure of Lead battery (铅蓄电池结构)

3 or 6 single cells , the single voltage is about 2 v.

3或6只单格电池串联组成，单格电压2V左右

Auto power : 12 v, 24 v or 6 v;

Electrolyte(电解液);

Electrolyte-adding hole(加液孔), air vent(通气孔)

② Capacity of the battery (蓄电池的容量)

20 hours rated capacity(20小时额定容量) (C20), unit : Ah;

rated Reserve capacity (Cm)(额定储备容量),unit: min;

cold start capability, Is(冷起动容量) unit:A

2 Electric Generator(发电机)

Requirement:

- ◆ Output current is DC(直流);
- ◆ Constant terminal voltage(端电压恒定), battery voltage $\leq$ output voltage $\leq$ electrical equipment permissible voltage;
- ◆ In high speed, the quantity of electricity can meet most of the electricity equipment needs.

Types of the silicon rectifying generator based on overall structure

- ◆ Conventional AC generator (普通交流发电机);
- ◆ Integrate alternator (整体式交流发电机);
- ◆ Brushless alternator (无刷交流发电机)
- ◆ Permanent magnet alternator (永磁交流发电机).

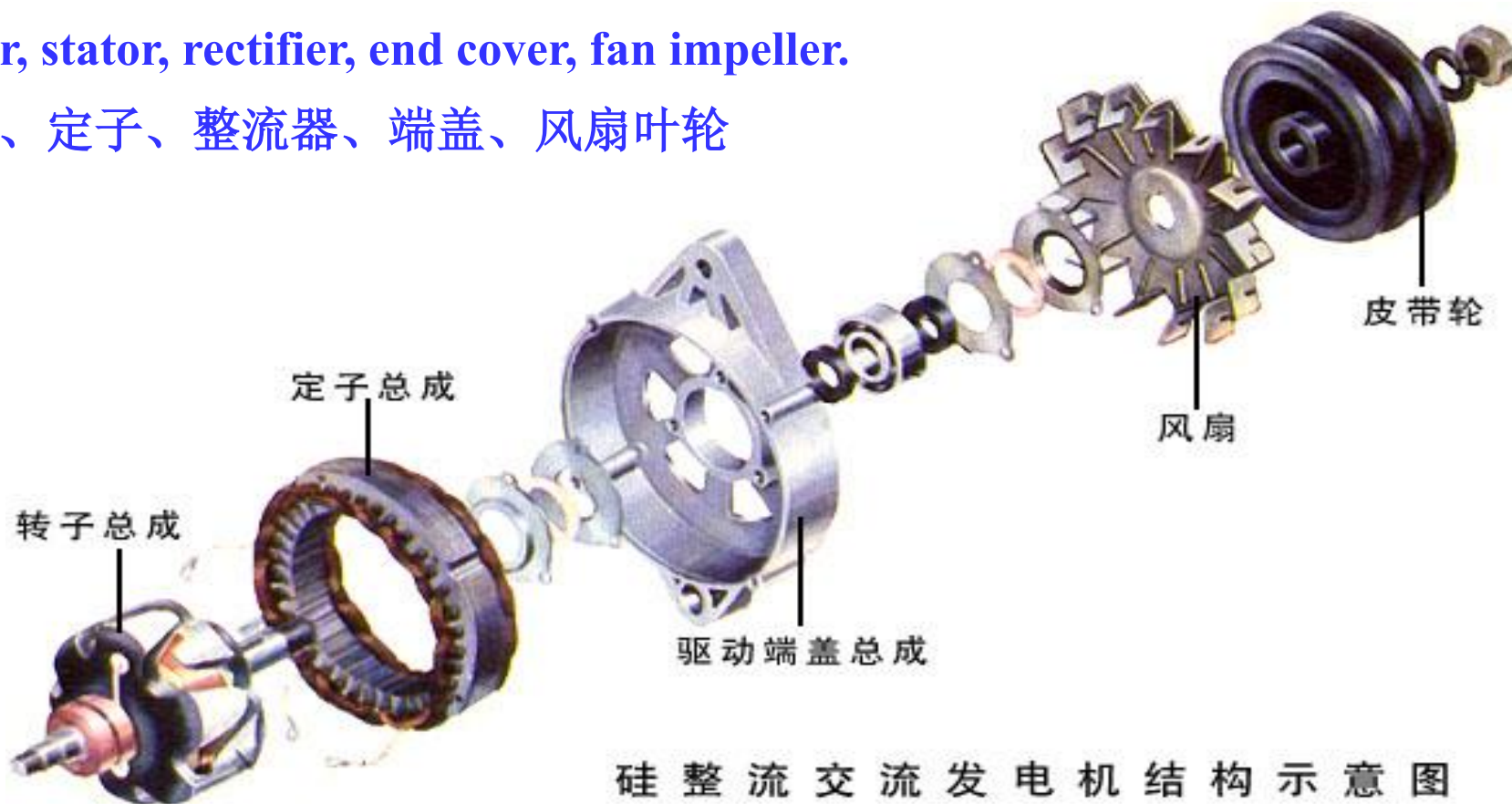
Structure of the silicon rectification alternator

(硅整流交流发电机的结构)

◆composition

Rotor, stator, rectifier, end cover, fan impeller.

转子、定子、整流器、端盖、风扇叶轮

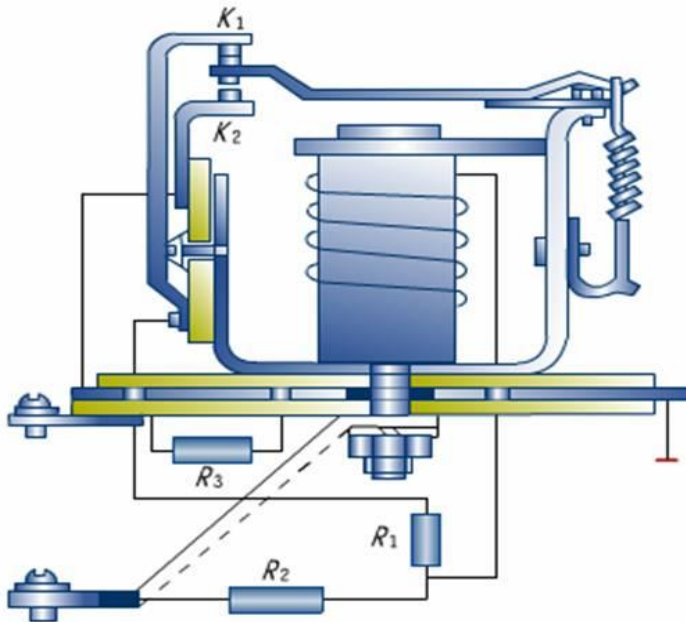


硅整流交流发电机结构示意图

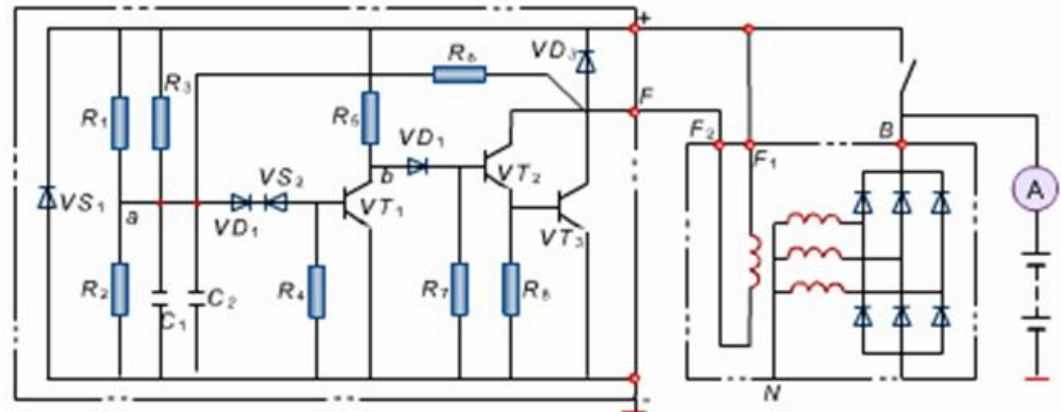
3 Voltage regulator(调节器)

◆Function:

keep generator terminal voltage constant when generator's speed changes.



双级触点式电压调节器结构图



JFT-106型晶体管电压调节器电路原理图

Contact oscillating type voltage regulator
(触点振荡式电压调节器)

Transistor voltage regulator
(晶体管电压调节器)

Automotive 42 V power supply system (42V汽车供电系统)

The evolution of Automotive Power Supply

1920年代	1955年	2000年	20X0年
7V (蓄电池6V)	14V (蓄电池12V)	42V (蓄电池36V)	多电压电源
负载, 辅机 • 车灯 • 点火	负载, 车体电气 • 车灯, EFI • 音响, 导航	负载, 功率电气 • HV, ISA, by-wire • 燃油消耗、排放	FC (300V) 功率 (42V) ECU (6V)

Reasons for using 42 v

- In a range of safe voltage (considering voltage fluctuation 电压波动)
- Integral multiple (整数倍) the existing voltage (combination use of the 12 v battery)
- Directly use 60 v of power semiconductor devices.
- Need no electric shock protection (触电防护) measures, continue to use the existing wire
- The output power of motor : 1 to 10 kW

Influences of using 42 v

Engine:

no cam shaft in the electromagnetic gas phase mechanism(无凸轮轴的电
磁式电控配气相位机构)

The use of compound flywheel
starter----generator(飞轮复合式启动
——发电机的使用)

Vehicle (整车):

Electric brake system(电动制动系统)

Electric clutch system(电动离合器系统)

Electric power steering system(电动动力
转向系统)

Water pump and oil pump driven by
electricity system(电动水泵、油泵系统)